

# **D-ILA<sup>®</sup> Projector**

**DLA-RS4200, DLA-NZ900, DLA-N1188, DLA-V900R  
DLA-RS3200, DLA-NZ800, DLA-N988, DLA-V800R**

## **External Command Communication Specification**

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JVCKENWOOD Corp.  
Projector Division

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# 1 Outline

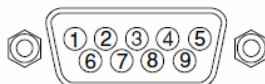
This specification describes how to control the D-ILA projector\* by using an external controller through the RS-232C interface.

## 2 Interface

### 2.1 Terminal

D-SUB 9pin Male terminal

| Pin No. | Name | Pin No. | Name |
|---------|------|---------|------|
| 1       | NC   | 6       | NC   |
| 2       | RXD  | 7       | NC   |
| 3       | TXD  | 8       | NC   |
| 4       | NC   | 9       | NC   |
| 5       | GND  |         |      |



### 2.2 External Controller Connector

Serial port connector (RS-232C)

For type of the connector and pin layout, please refer to each controller's specifications.

### 2.3 Communication Line

This control system uses RXD (receive data), TXD(transmit data) and GND line. Use an RS-232C crossover cable to connect the projector to the external controller like as PC.

## 3 Protocol

### 3.1 Communication specification

|                      |                      |
|----------------------|----------------------|
| Communication System | Asynchronous         |
| Interface            | RS-232C              |
| Baud rate            | 19200 bps            |
| Data length          | 8 bits               |
| Parity               | None                 |
| Stop bit             | 1 bit                |
| Flow control         | None                 |
| Communication code   | ASCII character code |

|  |    |    |    |    |    |    |    |    |    |    |  |
|--|----|----|----|----|----|----|----|----|----|----|--|
|  | ST | b0 | b1 | b2 | b3 | b4 | b5 | b6 | b7 | SP |  |
|--|----|----|----|----|----|----|----|----|----|----|--|

### 3.2 Data format

Control commands consist of Header, Unit ID, Command, Data and End. (Refer to the below)

\*The length of the control command varies according to function.

|        |         |  |         |  |           |     |         |        |
|--------|---------|--|---------|--|-----------|-----|---------|--------|
| 1 byte | 2 bytes |  | 2 bytes |  | n+1 bytes |     |         | 1 byte |
| Header | Unit ID |  | Command |  | Data[0]   | ... | Data[n] | End    |

Header: Indicates the start of communication (see paragraph 3.3, Header table).

Unit ID: Specifies the device to be controlled.

Command: See paragraph 3.5, Command table.

Data [i]: Parameter corresponding to the command (data i = 0, 1, .., n).

End: Indicates the end of communication.

### 3.3 Header table

Added header varies according to type of control command.

| HEX  | ASCII | Type              |
|------|-------|-------------------|
| 0x21 | '!    | Operation command |
| 0x3F | '?    | Reference command |
| 0x40 | '@'   | Response command  |
| 0x06 | ACK   | ACK               |

Operation command: Added when there is an operation command notification.

Reference command: Added when there is a reference command notification.

Response command: Added when there is a response command notice in response to a reference.

ACK: ACK response is given if the command reception is normal.

### 3.4 Unit ID table

- ♦ The unit ID consists of two bytes, the unit code and the individual code.
- ♦ The unit code is 0x89(Fixed)
- ♦ The individual code signifies "projector ID" and it is fixed on 0x01.
- ♦ A changed individual code is preserved on the unit side.

| Unit code |    |    |    |    |    |    |    |       |
|-----------|----|----|----|----|----|----|----|-------|
| b7        | b6 | b5 | b4 | b3 | b2 | b1 | b0 |       |
| 1         | 0  | 0  | 0  | 1  | 0  | 0  | 1  | Fixed |

| Individual code |    |    |    |    |    |    |    |                |
|-----------------|----|----|----|----|----|----|----|----------------|
| b7              | b6 | b5 | b4 | b3 | b2 | b1 | b0 |                |
| 0               | 0  | 0  | 0  | 0  | 0  | 0  | 1  | ID = 1 (Fixed) |

### 3.5 Command table

The command consists of two bytes of ASCII characters.

| HEX   |      | ASCII |     | Function                                                           | Operation | Reference |
|-------|------|-------|-----|--------------------------------------------------------------------|-----------|-----------|
| 0x00, | 0x00 | NUL   | NUL | NULL command                                                       | ✓         | -         |
| 0x50, | 0x57 | 'P'   | 'W' | Power [PoWer]                                                      | ✓         | ✓         |
| 0x49, | 0x50 | 'I'   | 'P' | Input [InPut]                                                      | ✓         | ✓         |
| 0x52, | 0x43 | 'R'   | 'C' | Remote control code through [Remote Code]                          | ✓         | -         |
| 0x53, | 0x55 | 'S'   | 'U' | Initial setup [SetUp]                                              | ✓         | ✓         |
| 0x47, | 0x52 | 'G'   | 'R' | Gamma data (Red) of the Gamma table "Custom 1/2/3" [Gamma Red]     | ✓         | ✓         |
| 0x47, | 0x47 | 'G'   | 'G' | Gamma data (Green) of the Gamma table "Custom 1/2/3" [Gamma Green] | ✓         | ✓         |
| 0x47, | 0x42 | 'G'   | 'B' | Gamma data (Blue) of the Gamma table "Custom 1/2/3" [Gamma Blue]   | ✓         | ✓         |
| 0x50, | 0x52 | 'P'   | 'R' | Red of Panel Alignment (zone)                                      | ✓         | ✓         |
| 0x50, | 0x42 | 'P'   | 'B' | Blue of Panel Alignment (zone)                                     | ✓         | ✓         |
| 0x53, | 0x43 | 'S'   | 'C' | Source asking [SourCe]                                             | -         | ✓         |
| 0x4D, | 0x44 | 'M'   | 'D' | Model status asking [MoDe]                                         | -         | ✓         |
| 0x50, | 0x4D | 'P'   | 'M' | Picture adjustment [adjustment of Picture] : Picture Adjust        | ✓         | ✓         |
| 0x49, | 0x53 | 'I'   | 'S' | Picture adjustment [adjustment of Picture] : Input Signal          | ✓         | ✓         |
| 0x49, | 0x4E | 'I'   | 'N' | Picture adjustment [adjustment of Picture] : Installation          | ✓         | ✓         |
| 0x44, | 0x53 | 'D'   | 'S' | Picture adjustment [adjustment of Picture] : Display Setup         | ✓         | ✓         |
| 0x46, | 0x55 | 'F'   | 'U' | Picture adjustment [adjustment of Picture] : Function              | ✓         | ✓         |
| 0x49, | 0x46 | 'I'   | 'F' | Picture adjustment [adjustment of Picture] : Information           | -         | ✓         |
| 0x4C, | 0x53 | 'L'   | 'S' | LAN setup [Lan Setup]                                              | ✓         | ✓         |
| 0x53, | 0x53 | 'S'   | 'S' | Service setup [Service Setup]                                      | ✓         | ✓         |

## 3.6 Parameter

### 3.6.1 Numeric value parameters

Signed 2-byte hexadecimal code represented by 4 (byte) characters.

Ex-1)

The parameter indication '20' (decimal):

Since '20' (decimal) is represented as '0014' in signed 2-byte hexadecimal, its parameter is:

'0014'(30H 30H 31H 34H)

Ex-2)

The parameter to indicate '-2' (decimal):

Since '-2' (decimal) is represented as 'FFFE' in signed 2-byte hexadecimal, its parameter is:

'FFFE'(46H 46H 46H 45H)

### 3.6.2 Special parameter

The parameters are generally interpreted with ASCII characters.

But some of the commands have a unique interpretation (for the details, see the section on Command sequences).

| HEX         | ASCII   | Meaning        |
|-------------|---------|----------------|
| 0x2B        | '+'     | '+'            |
| 0x2D        | '-'     | '-'            |
| 0x30        | '0'     | OFF/NO/Disable |
| 0x31        | '1'     | ON/YES/Enable  |
| 0x30 ~ 0x39 | '0'~'9' | '0'~'9'        |
| 0x41 ~ 0x5A | 'A'~'Z' | 'A'~'Z'        |



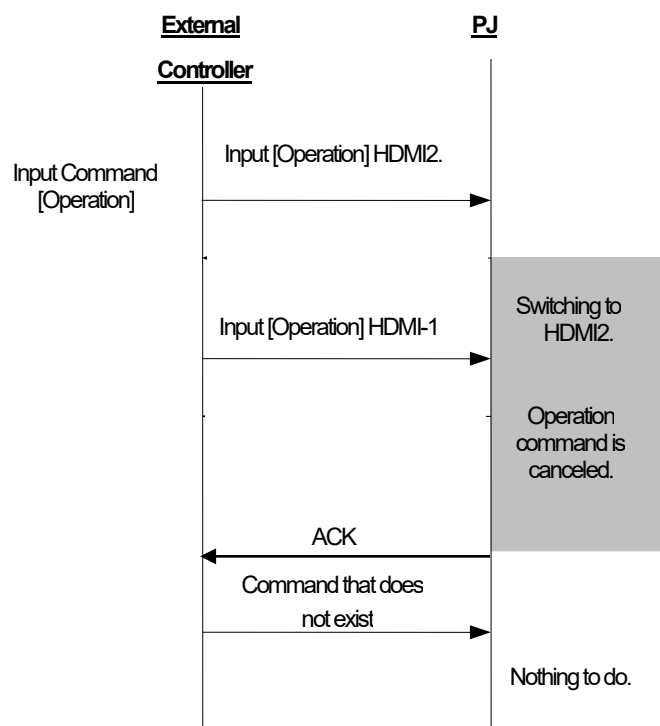
### 3.7 Exit code

0x0A(LineFeed) fixed.

### 3.8 Error handling

- ◆ An external controller should not transmit the next commands until it receives an ACK that the transmitted unit ID and the command match.
- ◆ The specifications for timeout and retry when an ACK response has not been received are not specified here; the specifications unique to the external controller may be used.
- ◆ If the byte interval is blank for 50 ms or longer, initialize the transmit-receive sequence (the received data is discarded).
- ◆ If a unit ID other than its own is received, it is ignored.
- ◆ If the unit ID matches but an undefined header/command is received, it is ignored.
- ◆ If the header/command is normal but an undefined parameter is received, it is ignored.
- ◆ Even if the command receipt is normal (ACK response) and the parameter is valid, it may be ignored, depending on the state of the projector.  
For the details, see the instruction manual for the projector (for example, projector, power ON operation in the power cooling state, etc.).

### 3.9 Communication sequence

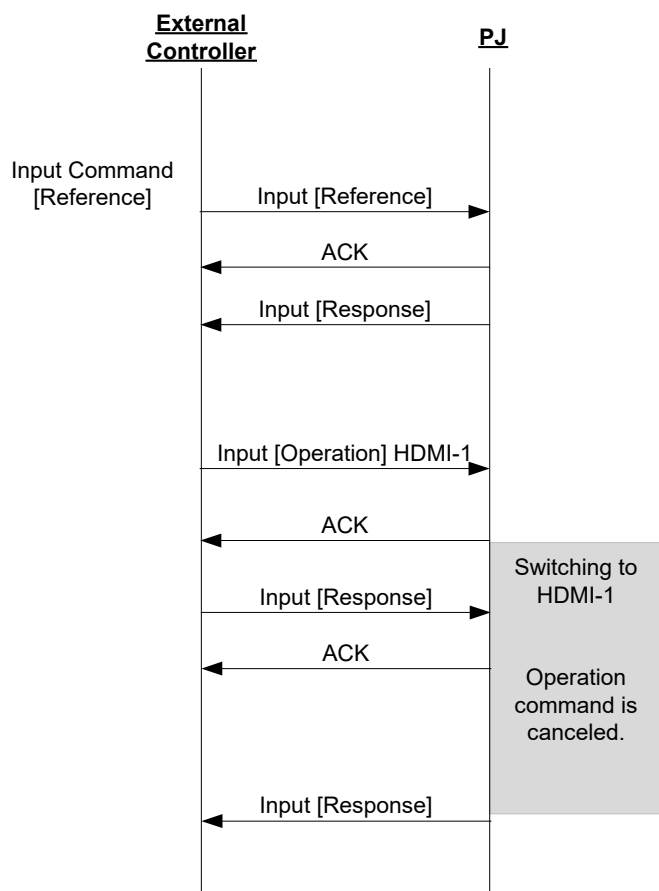


#### [ACK/NACK concept]

When the projector receives a command, it returns ACK after performs if the command is prescribed. If not, the projector returns nothing.

#### [Operation sequence]

When projector receives an operation command, it returns ACK after performs the command operation. However, if the projector receives the command while user operation is going on, the projector returns nothing.



#### [Reference sequence]

When projector receives a reference command, it returns ACK immediately, and then transmits related information for command as response to external controller.

If the projector receives a reference command while in user operation, it transmits ACK immediately, and then it transmits response command after the completion of user operation.

This is because of conflict between projector information and the information passed to external controller via response command based on timing, if the reference command is approved while in user operation.

## 4 Command control

### 4.1 NULL command

In using CEDIA commands, it is used in order to confirm whether transmission is possible.

It is used to confirm whether the external controller and the projector are properly connected.

#### Command code

Table 4-1 NULL CMD

| HEX  |      | ASCII |      | Function     | Operation | Reference |
|------|------|-------|------|--------------|-----------|-----------|
| 0x00 | 0x00 | NULL  | NULL | NULL command | ✓         | -         |

#### Parameter

None

#### 4.1.1 Operation

Use the command as shown in the following examples.

**Purpose: To confirm whether the external controller and the projector are properly connected.**

- (1) Data is transmitted from the external controller to the projector as follows.

| 1: External controller → |      |              |      |      |      |
|--------------------------|------|--------------|------|------|------|
| 0x21                     | 0x89 | 0x01         | 0x00 | 0x00 | 0x0A |
| !(Operation)             | PJ   | Individual:1 | NULL | NULL | End  |

- (2) If the external controller and the projector are connected and the projector is operating normally, the projector returns an ACK to the external controller as a NULL command response.

| 2: ← Projector |      |                |      |      |      |
|----------------|------|----------------|------|------|------|
| 0x06           | 0x89 | 0x01           | 0x00 | 0x00 | 0x0A |
| ACK            | PJ   | Individual : 1 | NULL | NULL | End  |

By the above exchange, it can be confirmed whether the connection and communication are normal or abnormal.

#### 4.1.2 Reference

N/A

## 4.2 Power [PoWer]

Used for power ON/OFF operation and for referencing the power setting state of the projector.

### Command code

Table 4-2 POWER CMD

| HEX  |      | ASCII |     | Function      | Operation | Reference |
|------|------|-------|-----|---------------|-----------|-----------|
| 0x50 | 0x57 | 'P'   | 'W' | Power [POWER] | ✓         | ✓         |

### Parameters

Table 4-3 POWER CMD DATA

| HEX  | ASCII | Operation |
|------|-------|-----------|
| 0x30 | '0'   | Power OFF |
| 0x31 | '1'   | Power ON  |

【Data 0】

Table 4-4 POWER CMD STATUS

| HEX  | ASCII | Operation            |
|------|-------|----------------------|
| 0x30 | '0'   | Standby              |
| 0x31 | '1'   | Power On             |
| 0x32 | '2'   | Cooling              |
| 0x33 | '3'   | Sequence of power-on |
| 0x34 | '4'   | Emergency            |

### 4.2.1 Operation

Use the command as shown in the following examples.

#### Purpose: To turn the Projector's power OFF. (Current state: Power-ON)

- (1) Data is transmitted from the external controller to the Projector as follows.

| 1: External controller → |      |                |      |      |      |      |
|--------------------------|------|----------------|------|------|------|------|
| 0x21                     | 0x89 | 0x01           | 0x50 | 0x57 | 0x30 | 0x0A |
| !(Operation)             | PJ   | Individual : 1 | 'P'  | 'W'  | OFF  | End  |

- (2) If the projector receives data (1) and the command reception is normal, the projector returns an ACK as follows.

| 2: ← Projector |      |                |      |      |      |
|----------------|------|----------------|------|------|------|
| 0x06           | 0x89 | 0x01           | 0x50 | 0x57 | 0x0A |
| ACK            | PJ   | Individual : 1 | 'P'  | 'W'  | End  |

- (3) The projector turns power OFF.

The power can be turned OFF by the above exchange.

- The power-OFF operation can be done by a sequence similar to that for power-ON.
- The projector ignores data in the same state. For example, even if power-ON data is sent with the projector in power-ON mode, no projector operation is done.

## 4.2.2 Reference

Use the command as shown in the following examples.

### Purpose: To confirm the present projector's power state. (Current state: Power-ON)

- (1) Data is transmitted from the external controller to the projector as follows.

| 1: External controller → |      |                |      |      |      |
|--------------------------|------|----------------|------|------|------|
| 0x3F                     | 0x89 | 0x01           | 0x50 | 0x57 | 0x0A |
| '?' (Reference)          | PJ   | Individual : 1 | 'P'  | 'W'  | End  |

- (2) If the projector receives data (1) above and the command reception was normal, the projector returns an ACK as follows.

| 2: ← Projector |      |                |      |      |      |
|----------------|------|----------------|------|------|------|
| 0x06           | 0x89 | 0x01           | 0x50 | 0x57 | 0x0A |
| ACK            | PJ   | Individual : 1 | 'P'  | 'W'  | End  |

- (3) Next, the projector transmits a report of the power setting to the external controller.

| 3: ← Projector |      |                |      |      |      |      |
|----------------|------|----------------|------|------|------|------|
| 0x40           | 0x89 | 0x01           | 0x50 | 0x57 | 0x31 | 0x0A |
| '@'(Response)  | PJ   | Individual : 1 | 'P'  | 'W'  | ON   | End  |

By the above exchange, it can be confirmed that the projector's power state is ON.

Operation on projector screen

None

### 4.3 Input [InPut]

Used for input switching operation and referencing the input settings of the projector.

Setting is automatically stored in the projector when input is changed.

#### Command code

Table 4-5 INPUT CMD

| HEX  |      | ASCII |     | Function             | Operation | Reference |
|------|------|-------|-----|----------------------|-----------|-----------|
| 0x49 | 0x50 | 'I'   | 'P' | Input switch [INPUT] | ✓         | ✓         |

#### Parameters

Data length: 1 or 2

【Data 0】

Table 4-6 INPUT CMD DATA

| HEX  | ASCII | Operation |
|------|-------|-----------|
| 0x36 | '6'   | HDMI-1    |
| 0x37 | '7'   | HDMI-2    |

- ♦ The input switching operation is not done if the parameter data is sent to a terminal that is not provided on the projector.
- ♦ The toggle sequence follows the sequence of Operation panel on the projector.

### 4.3.1 Operation

Use the command as shown in the following examples.

#### Purpose: To switch the input to HDMI2. (Current input state: "HDMI-1")

- (1) Data is transmitted from the external controller to the projector as follows.

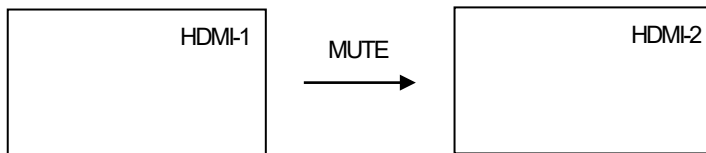
| 1: External controller → |      |                |      |      |        |      |
|--------------------------|------|----------------|------|------|--------|------|
| 0x21                     | 0x89 | 0x01           | 0x49 | 0x50 | 0x37   | 0x0A |
| '!(Operation)            | PJ   | Individual : 1 | 'I'  | 'P'  | HDMI-2 | End  |

- (2) If projector receives data (1) and the command receipt is normal, an ACK is returned from the projector as follows.

| 2: ← Projector |      |                |      |      |      |
|----------------|------|----------------|------|------|------|
| 0x06           | 0x89 | 0x01           | 0x49 | 0x50 | 0x0A |
| ACK            | PJ   | Individual : 1 | 'I'  | 'P'  | End  |

- (3) The projector switches the input to HDMI-2.

Operation on the projector screen



### 4.3.2 Reference

Use the command as shown in the following examples.

#### Purpose: To confirm the current projector output. (Current projector state: HDMI-2)

- (1) Transmit the data from the external controller to the projector as follows.

| 1: External controller → |      |                |      |      |      |
|--------------------------|------|----------------|------|------|------|
| 0x3F                     | 0x89 | 0x01           | 0x49 | 0x50 | 0x0A |
| '?(Reference)            | PJ   | Individual : 1 | 'I'  | 'P'  | End  |

- (2) If the projector receives the data (1) above and the command receipt is normal, an ACK is returned from the projector as follows.

| 2: ← Projector |      |                |      |      |      |
|----------------|------|----------------|------|------|------|
| 0x06           | 0x89 | 0x01           | 0x49 | 0x50 | 0x0A |
| ACK            | PJ   | Individual : 1 | 'I'  | 'P'  | End  |

- (3) The projector transmits information of input setting to the external controller.

| 3: ← Projector |      |                |      |      |        |      |
|----------------|------|----------------|------|------|--------|------|
| 0x40           | 0x89 | 0x01           | 0x49 | 0x50 | 0x37   | 0x0A |
| '@(Response)   | PJ   | Individual : 1 | 'I'  | 'P'  | HDMI-2 | End  |

By the above exchange, it can be confirmed that the projector output state is in the HDMI-2 state.

Operation on the projector screen

None

## 4.4 Remote control pass-through [RemoteCode]

By selecting JVC remote control code, the same operation as user remote control can be achieved.

Remote control code consists of one byte of custom code and one byte of function/operation code.

Remote control code varies according to the projector and its state.

For the details of the remote control codes, see the key code specifications of each model.

### Command code

Table 4-7 REMO CMD

| HEX  |      | ASCII |     | Function                                  | Operation | Reference |
|------|------|-------|-----|-------------------------------------------|-----------|-----------|
| 0x52 | 0x43 | 'R'   | 'C' | Remote control pass-through [Remote Code] | ✓         |           |

### Parameters

Data length: 4

Table 4-8 REMO CMD DATA

| HEX         | ASCII     | Operation                   |
|-------------|-----------|-----------------------------|
| 0x30 ~ 0x39 | '0' ~ '9' | Remote control code setting |
| 0x41 ~ 0x46 | 'A' ~ 'F' | Remote control code setting |

- ◆ The remote control code specification is in hexadecimal digits.
- ◆ The operation transition with the remote control codes is the same as from the user remote control.
- ◆ For the details of the remote control codes, see the key code specifications.



#### 4.4.1 Operation

Use the command as shown in the following examples.

**Purpose: To display MENU screen by press the “MENU” of remote control code [0x732E].**

- (1) Transmit the data from the external controller to the projector as follows.

| 1: External controller → |      |                |      |      |      |      |      |      |      |
|--------------------------|------|----------------|------|------|------|------|------|------|------|
| 0x21                     | 0x89 | 0x01           | 0x52 | 0x43 | 0x37 | 0x33 | 0x32 | 0x45 | 0x0A |
| !(Operation)             | PJ   | Individual : 1 | 'R'  | 'C'  | '7'  | '3'  | '2'  | 'E'  | End  |

- (2) If the projector receives the data (1) above and the command receipt is normal, an ACK is returned from the projector as follows.

| 2: ← Projector |      |                |      |      |      |
|----------------|------|----------------|------|------|------|
| 0x06           | 0x89 | 0x01           | 0x52 | 0x43 | 0x0A |
| ACK            | PJ   | Individual : 1 | 'R'  | 'C'  | End  |

- (3) The projector produces a MENU screen.

By the above exchange, the MENU screen can be produced.

## 4.5 Setup [SetUp]

Used to change the initial setting.

### Command code

Table 4-9 SETUP CMD

| HEX  |       | ASCII |     | Function                |
|------|-------|-------|-----|-------------------------|
| 0x53 | 0X55R | 'S'   | 'U' | Initial setting [SetUp] |

### Parameters

Data length: No regulation

Sub command table (Mandatory command only)

Table 4-10 SETUP CMD SUB

| HEX  |      | ASCII |     | Function                                     | Last Memory | Operation | Reference |
|------|------|-------|-----|----------------------------------------------|-------------|-----------|-----------|
| 0x52 | 0x53 | 'R'   | 'S' | Switch the external control command protocol | No          | ✓         | -         |
| 0x52 | 0x43 | 'R'   | 'C' | Switch the IR code                           | No          | ✓         | ✓         |

- ◆ It consists of "Sub command" + "Setting". The Sub command consists of ASCII character two bytes.

Sub commands parameters are as follow.

Parameters when the Sub command is [0x52,0x53] : Data length 1

| HEX  | ASCII | Operation                 |
|------|-------|---------------------------|
| 0x31 | '1'   | Compatible command system |

Parameters when the Sub command is [0x52,0x43] : Data length 1

| HEX  | ASCII | Operation    |
|------|-------|--------------|
| 0x30 | '0'   | A code(0x73) |
| 0x31 | '1'   | B code(0x63) |

### 4.5.1 Operation

Use the command as shown in the following examples.

#### **Purpose: To switch External control command to compatible command protocol.**

- (1) Transmit the data from the external controller to the projector as follows.

| <b>1: External controller →</b> |      |                |      |      |      |      |                                  |      |
|---------------------------------|------|----------------|------|------|------|------|----------------------------------|------|
| 0x21                            | 0x89 | 0x01           | 0x53 | 0x55 | 0x52 | 0x53 | 0x31                             | 0x0A |
| !(Operation)                    | PJ   | Individual : 1 | 'S'  | 'U'  | 'R'  | 'S'  | Compatible<br>command protocol:1 | End  |

- (2) If the projector receives the data (1) above and the command receipt is normal, an ACK is returned from the projector as follows.

| <b>2: ← Projector</b> |      |                |      |      |      |
|-----------------------|------|----------------|------|------|------|
| 0x06                  | 0x89 | 0x01           | 0x53 | 0x55 | 0x0A |
| ACK                   | PJ   | Individual : 1 | 'S'  | 'U'  | End  |

- (3) The projector switches external command protocol to Compatible command protocol.

By the above exchange, the projector switches to Compatible command protocol.

Operation on projector screen

None

## 4.6 Gamma data of Gamma table “Custom 1/2/3” [GammaRed, Green, Blue]

Used for sending operation of the gamma data when the Gamma table is Custom, and referencing the gamma data of the projector.

Target of switching and referencing is followed by setting value of the Gamma bank.

### Command code

Table 4-11 GAMMA DATA CMD

| HEX  |      | ASCII |     | Function                                                            | Operation | Reference |
|------|------|-------|-----|---------------------------------------------------------------------|-----------|-----------|
| 0x47 | 0x52 | 'G'   | 'R' | Gamma data of the gamma table "Custom1/2/3" (Red)<br>[GammaRed]     | ✓         | ✓         |
| 0x47 | 0x47 | 'G'   | 'G' | Gamma data of the gamma table "Custom1/2/3"<br>(Green) [GammaGreen] | ✓         | ✓         |
| 0x47 | 0x42 | 'G'   | 'B' | Gamma data of the gamma table "Custom1/2/3" (Blue)<br>[GammaBlue]   | ✓         | ✓         |

### Parameters

Data length: 512

The gamma data has 256 adjustment points composed of binary data.

The byte order is little endian.

- When the projector is not powered on and it receives the command, the projector ignores the command and does not respond.
- When the gamma table of the projector is not "Custom1", "Custom2" or "Custom3" and the projector receives the command, the projector ignores the command and does not respond.
- When the projector receives a gamma data by operation command, the projector saves the gamma data on proper area of EEPROM immediately.
- If the projector receives gamma data larger than 512 bytes, it is judged as invalid data and the projector does not respond.
- The curve is combination of the curve on the OSD and the table (Normal/A/B/C/D) in the video processor.

#### 4.6.1 Operation

Use the command as shown in the following examples.

**Purpose: To send red gamma data of the gamma table “Custom1/2/3” to the projector.**

- (1) Transmit the data from the external controller to the projector as follows.

| 1: External controller → |      |                |      |      |      |
|--------------------------|------|----------------|------|------|------|
| 0x21                     | 0x89 | 0x01           | 0x47 | 0x52 | 0x0A |
| !(Operation)             | PJ   | Individual : 1 | 'G'  | 'R'  | End  |

- (2) If the projector receives the data (1) above and the command receipt is normal, an ACK is returned from the projector as follows.

| 2: ← Projector |      |                |      |      |      |
|----------------|------|----------------|------|------|------|
| 0x06           | 0x89 | 0x01           | 0x47 | 0x52 | 0x0A |
| ACK            | PJ   | Individual : 1 | 'G'  | 'R'  | End  |

- (3) The external controller transmits 512 bytes of binary data to the projector.

- (4) If the projector receives data (3) above and the command receipt was normal, an ACK is returned from the projector as follows.

| 4: ← Projector |      |                |      |      |      |
|----------------|------|----------------|------|------|------|
| 0x06           | 0x89 | 0x01           | 0x47 | 0x52 | 0x0A |
| ACK            | PJ   | Individual : 1 | 'G'  | 'R'  | End  |

#### 4.6.2 Reference

Use the command as shown in the following examples.

**Purpose: To confirm the green gamma data of the current gamma table “Custom3”. (Current Gamma bank: Custom3)**

- (1) Transmit the data from the external controller to the projector as follows.

| 1: External controller → |      |                |      |      |      |
|--------------------------|------|----------------|------|------|------|
| 0x3F                     | 0x89 | 0x01           | 0x47 | 0x47 | 0x0A |
| ?(Reference)             | PJ   | Individual : 1 | 'G'  | 'G'  | End  |

- (2) If the projector receives the data (1) above and the command receipt is normal, an ACK is returned from the projector as follows.

| 2: ← Projector |      |                |      |      |      |
|----------------|------|----------------|------|------|------|
| 0x06           | 0x89 | 0x01           | 0x47 | 0x47 | 0x0A |
| ACK            | PJ   | Individual : 1 | 'G'  | 'G'  | End  |

- (3) The projector transmits 512 bytes binary data to the external controller.

### 4.7 Panel Alignment (zone) Data [Panel Alignment(Zone) Red, Blue]

Used for transmitting operation and referencing Panel Alignment (zone) data

## Command Code

Table 4-12 PANEL ALIGNMENT(ZONE) DATA CMD

| HEX  |      | ASCII |     | Function                                 | transmitting | referencing |
|------|------|-------|-----|------------------------------------------|--------------|-------------|
| 0x50 | 0x52 | 'P'   | 'R' | Data of Red [Panel Alignment(Zone)Red]   | ✓            | ✓           |
| 0x50 | 0x42 | 'P'   | 'B' | Data of Blue [Panel Alignment(Zone)Blue] | ✓            | ✓           |

## Parameters

Data Length: 256

Horizontal and Vertical data of 11x11 Adjustment zone composed of binary data. Data could be -31 (0xE1) to +31(0x1F).

Data is assigned by 2 bytes and its order is from Horizontal to Vertical. The order of Adjustment zone is shown at a table below. 121 (Adjustment zone) x 2 (Horizontal / Vertical) + 13 (reserved) = 256 Byte

| Data No. | Horizontal position of zone | Vertical position of zone | Horizontal / Vertical |
|----------|-----------------------------|---------------------------|-----------------------|
| 1        | 0                           | 0                         | Horizontal            |
| 2        |                             |                           | Vertical              |
| 3        | 1                           | 0                         | Horizontal            |
| 4        |                             |                           | Vertical              |
| 5        | 2                           | 0                         | Horizontal            |
| 6        |                             |                           | Vertical              |
| (skip)   |                             |                           |                       |
| 21       | 10                          | 0                         | Horizontal            |
| 22       |                             |                           | Vertical              |
| 23       | 0                           | 1                         | Horizontal            |
| 24       |                             |                           | Vertical              |
| (skip)   |                             |                           |                       |
| 239      | 9                           | 10                        | Horizontal            |
| 240      |                             |                           | Vertical              |
| 241      | 10                          | 10                        | Horizontal            |
| 242      |                             |                           | Vertical              |
| 243-256  | Reserved                    |                           |                       |

The Byte order is little endian.

- ◆ When the projector is not powered on and it receives the command, the projector ignores the command and does not respond.
- ◆ If the projector receives gamma data larger than 256 bytes, it is judged as invalid data and the projector does not respond.

### 4.7.1 Operation

Use the command as shown in the following examples.

#### Purpose : To send red Alignment (zone) data to the projector.

- (1) Transmit the data from the external controller to the projector as follows.

| 1: External Controller → |      |                |      |      |      |
|--------------------------|------|----------------|------|------|------|
| 0x21                     | 0x89 | 0x01           | 0x50 | 0x52 | 0x0A |
| '!' (Operation)          | PJ   | Individual : 1 | 'P'  | 'R'  | End  |

- (2) If the projector receives the data (1) above and the command receipt is normal, an ACK is returned from the projector as follows.

| 2: ← Projector |      |                |      |      |      |
|----------------|------|----------------|------|------|------|
| 0x06           | 0x89 | 0x01           | 0x50 | 0x52 | 0x0A |
| ACK            | PJ   | Individual : 1 | 'P'  | 'R'  | End  |

- (3) The external controller transmits 256 bytes of binary data to the projector.

- (4) If the projector receives data (3) above and the command receipt was normal, an ACK is returned from the projector as follows.

| 4: ← Projector |      |                |      |      |      |
|----------------|------|----------------|------|------|------|
| 0x06           | 0x89 | 0x01           | 0x50 | 0x52 | 0x0A |
| ACK            | PJ   | Individual : 1 | 'P'  | 'R'  | End  |

### 4.7.2 Reference

Use the command as shown in the following examples.

#### Purpose : To confirm Blue data of the current Panel Alignment (zone)

- (1) Transmit the data from the external controller to the projector as follows.

| 1: External Controller → |      |              |      |      |      |
|--------------------------|------|--------------|------|------|------|
| 0x3F                     | 0x89 | 0x01         | 0x50 | 0x42 | 0x0A |
| '?' (reference)          | PJ   | Individual:1 | 'P'  | 'B'  | End  |

- (2) If the projector receives the data (1) above and the command receipt is normal, an ACK is returned from the projector as follows.

| 2: ← Projector |      |                |      |      |      |
|----------------|------|----------------|------|------|------|
| 0x06           | 0x89 | 0x01           | 0x50 | 0x42 | 0x0A |
| ACK            | PJ   | Individual : 1 | 'P'  | 'B'  | End  |

- (3) The projector transmits 256 bytes binary data to the external controller.

## 4.8 Source Asking [SourCe]

Use to refer signal input status of the projector.

### Command code

Table 4-13 SOURCE CMD

| HEX  |      | ASCII |     | Function               | Operation | Reference |
|------|------|-------|-----|------------------------|-----------|-----------|
| 0x53 | 0x43 | 'S'   | 'C' | Source asking [SourCe] |           | ✓         |

### Parameters

Data length: 1

【Data 0】

Table 4-14 SOURCE CMD DATA

| HEX  | ASCII | Operation                                   |
|------|-------|---------------------------------------------|
| 0x30 | '0'   | No signal or out of range                   |
| 0x31 | '1'   | Available signal is input to the projector. |

- ◆ When the projector is not powered on and it receives the command, the projector ignores the command and does not respond.

#### 4.8.1 Reference

Use the command as shown in the following examples.

**Purpose: To confirm the current status of the projector. (Current status: No signal or out of range)**

- (1) Transmit the data from the external controller to the projector as follows.

| 1: External controller → |      |                |      |      |      |
|--------------------------|------|----------------|------|------|------|
| 0x3F                     | 0x89 | 0x01           | 0x53 | 0x43 | 0x0A |
| ?(Reference)             | PJ   | Individual : 1 | 'S'  | 'C'  | End  |

- (2) If the projector receives the data (1) above and the command receipt is normal, an ACK is returned from the projector as follows.

| 2: ← Projector |      |                |      |      |      |
|----------------|------|----------------|------|------|------|
| 0x06           | 0x89 | 0x01           | 0x53 | 0x43 | 0x0A |
| ACK            | PJ   | Individual : 1 | 'S'  | 'C'  | End  |

- (3) The projector transmits the status of the input signal to the external controller.

| 3: ← Projector |      |                |      |      |                           |      |
|----------------|------|----------------|------|------|---------------------------|------|
| 0x40           | 0x89 | 0x01           | 0x53 | 0x43 | 0x30                      | 0x0A |
| '@'(Response)  | PJ   | Individual : 1 | 'S'  | 'C'  | No signal or out of range | End  |

By the above exchange, it can be confirmed that whether the status of input signal of the projector is no signal or out of range.



## 4.9 Model status asking [MoDel]

Used for referring model status of the projector.

### Command code

Table 4-15 MODEL STATUS CMD

| HEX  |      | ASCII |     | Function                    | Operation | Reference |
|------|------|-------|-----|-----------------------------|-----------|-----------|
| 0x4D | 0x44 | 'M'   | 'D' | Model status asking [MoDel] | -         | ✓         |

### Parameters

Data length: 14

Table 4-16 MODEL STATUS CMD DATA

| Parameters |      |      |      |      |      |      |      |      |      |      |      |      |        |
|------------|------|------|------|------|------|------|------|------|------|------|------|------|--------|
| 0x49       | 0x4C | 0x41 | 0x46 | 0x50 | 0x4A | 0x20 | 0x2D | 0x2D | 0x20 | 0x42 | 0x38 | 0x41 | (*1)   |
| 'I'        | 'L'  | 'A'  | 'F'  | 'P'  | 'J'  | SP   | '-'  | '-'  | SP   | 'B'  | '8'  | 'A'  | '(*1)' |

(\*1) DLA-NZ900 etc.. = '1'(0x31), DLA-NZ800 etc... = '2'(0x32)

### 4.9.1 Reference

Use the command as shown in the following examples.

#### Purpose: To confirm the model status of the current projector.

- (1) Transmit the data from the external controller to the projector as follows.

| 1: External controller → |      |                |      |      |      |
|--------------------------|------|----------------|------|------|------|
| 0x3F                     | 0x89 | 0x01           | 0x4D | 0x44 | 0x0A |
| '?(Reference)            | PJ   | Individual : 1 | 'M'  | 'D'  | End  |

- (2) If the projector receives the data (1) above and the command receipt is normal, an ACK is returned from the projector as follows.

| 2: ← Projector |      |                |      |      |      |
|----------------|------|----------------|------|------|------|
| 0x06           | 0x89 | 0x01           | 0x4D | 0x44 | 0x0A |
| ACK            | PJ   | Individual : 1 | 'M'  | 'D'  | End  |

- (3) The projector transmits a report of the model status to the external controller.

| 3: ← Projector |      |                |      |      |                     |      |
|----------------|------|----------------|------|------|---------------------|------|
| 0x40           | 0x89 | 0x01           | 0x4D | 0x44 | Parameters 14 bytes | 0x0A |
| '@(Response)   | PJ   | Individual : 1 | 'M'  | 'D'  | Model status        | End  |

By the above exchange, it can be confirmed model status of the projector.

## 4.10 Adjustment [Adjustment Command]

Used for adjusting some function.

### Command code

Table 4-17 ADJUSTMENT CMD

| HEX  |       | ASCII |     | Function       |
|------|-------|-------|-----|----------------|
| 0x50 | 0x4D  | 'P'   | 'M' | Picture Adjust |
| 0x49 | 0x53  | 'I'   | 'S' | Input Signal   |
| 0x49 | 0x4E  | 'I'   | 'N' | Installation   |
| 0x44 | 0x53  | 'D'   | 'S' | Display Setup  |
| 0x46 | 0X55R | 'F'   | 'U' | Function       |
| 0x49 | 0x46  | 'I'   | 'F' | Information    |

### Parameter1

Parameter1: Sub command

Data length: ASCII character 2 bytes

Parameter1 table is as follow.

Table 4-18 ADJUSTMENT CMD SUB

| Command Code |     | Parameter1 |     | Function                                    | Operation | Reference | Data type | Model        |              |
|--------------|-----|------------|-----|---------------------------------------------|-----------|-----------|-----------|--------------|--------------|
|              |     |            |     |                                             |           |           |           | NZ900 etc... | NZ800 etc... |
| 'P'          | 'M' | 'P'        | 'M' | Picture Mode switch                         | ✓         | ✓         | Special 4 | ✓            | ✓            |
| 'P'          | 'M' | 'P'        | 'R' | Color Profile switch (*1)                   | ✓         | ✓         | Special   | ✓            | ✓            |
| 'P'          | 'M' | 'C'        | 'L' | Color Temperature table switch              | ✓         | ✓         | Special   | ✓            | ✓            |
| 'P'          | 'M' | 'C'        | 'C' | Color Temperature Correction switch         | ✓         | ✓         | Special   | ✓            | ✓            |
| 'P'          | 'M' | 'G'        | 'R' | Color Temperature Gain (Red) adjustment     | ✓         | ✓         | Numeric   | ✓            | ✓            |
| 'P'          | 'M' | 'G'        | 'G' | Color Temperature Gain (Green) adjustment   | ✓         | ✓         | Numeric   | ✓            | ✓            |
| 'P'          | 'M' | 'G'        | 'B' | Color Temperature Gain (Blue) adjustment    | ✓         | ✓         | Numeric   | ✓            | ✓            |
| 'P'          | 'M' | 'O'        | 'R' | Color Temperature Offset (Red) adjustment   | ✓         | ✓         | Numeric   | ✓            | ✓            |
| 'P'          | 'M' | 'O'        | 'G' | Color Temperature Offset (Green) adjustment | ✓         | ✓         | Numeric   | ✓            | ✓            |
| 'P'          | 'M' | 'O'        | 'B' | Color Temperature Offset (Blue) adjustment  | ✓         | ✓         | Numeric   | ✓            | ✓            |
| 'P'          | 'M' | 'G'        | 'T' | Gamma Table switch                          | ✓         | ✓         | Special   | ✓            | ✓            |
| 'P'          | 'M' | 'F'        | 'W' | Picture Tone (White) adjustment             | ✓         | ✓         | Numeric   | ✓            | ✓            |
| 'P'          | 'M' | 'F'        | 'R' | Picture Tone (Red) adjustment               | ✓         | ✓         | Numeric   | ✓            | ✓            |
| 'P'          | 'M' | 'F'        | 'G' | Picture Tone (Green) adjustment             | ✓         | ✓         | Numeric   | ✓            | ✓            |
| 'P'          | 'M' | 'F'        | 'B' | Picture Tone (Blue) adjustment              | ✓         | ✓         | Numeric   | ✓            | ✓            |
| 'P'          | 'M' | 'C'        | 'N' | Contrast adjustment                         | ✓         | ✓         | Numeric   | ✓            | ✓            |
| 'P'          | 'M' | 'B'        | 'R' | Brightness adjustment                       | ✓         | ✓         | Numeric   | ✓            | ✓            |
| 'P'          | 'M' | 'C'        | 'O' | Color adjustment                            | ✓         | ✓         | Numeric   | ✓            | ✓            |
| 'P'          | 'M' | 'T'        | 'I' | Tint adjustment                             | ✓         | ✓         | Numeric   | ✓            | ✓            |
| 'P'          | 'M' | 'D'        | 'C' | Dynamic CTRL switch                         | ✓         | ✓         | Special   | ✓            | ✓            |
| 'P'          | 'M' | 'R'        | 'N' | NR adjustment                               | ✓         | ✓         | Numeric   | ✓            | ✓            |
| 'P'          | 'M' | 'G'        | 'C' | Gamma Correction switch                     | ✓         | ✓         | Special 4 | ✓            | ✓            |
| 'P'          | 'M' | 'D'        | 'R' | Gamma Red data                              | ✓         | ✓         | Special 2 | ✓            | ✓            |
| 'P'          | 'M' | 'D'        | 'G' | Gamma Green data                            | ✓         | ✓         | Special 2 | ✓            | ✓            |
| 'P'          | 'M' | 'D'        | 'B' | Gamma Blue data                             | ✓         | ✓         | Special 2 | ✓            | ✓            |
| 'P'          | 'M' | 'R'        | 'W' | Bright Level White                          | ✓         | ✓         | Numeric   | ✓            | ✓            |
| 'P'          | 'M' | 'R'        | 'R' | Bright Level Red                            | ✓         | ✓         | Numeric   | ✓            | ✓            |
| 'P'          | 'M' | 'R'        | 'G' | Bright Level Green                          | ✓         | ✓         | Numeric   | ✓            | ✓            |
| 'P'          | 'M' | 'R'        | 'B' | Bright Level Blue                           | ✓         | ✓         | Numeric   | ✓            | ✓            |
| 'P'          | 'M' | 'K'        | 'W' | Dark Level White                            | ✓         | ✓         | Numeric   | ✓            | ✓            |
| 'P'          | 'M' | 'K'        | 'R' | Dark Level Red                              | ✓         | ✓         | Numeric   | ✓            | ✓            |
| 'P'          | 'M' | 'K'        | 'G' | Dark Level Green                            | ✓         | ✓         | Numeric   | ✓            | ✓            |

|   |   |   |   |                                         |      |   |            |   |   |
|---|---|---|---|-----------------------------------------|------|---|------------|---|---|
| P | M | K | B | Dark Level Blue                         | ✓    | ✓ | Numeric    | ✓ | ✓ |
| P | M | C | B | Color Management table                  | ✓    | ✓ | Special    | ✓ | ✓ |
| P | M | H | R | HUE (Red) adjustment                    | ✓    | ✓ | Numeric    | ✓ | ✓ |
| P | M | H | Y | HUE (Yellow) adjustment                 | ✓    | ✓ | Numeric    | ✓ | ✓ |
| P | M | H | G | HUE (Green) adjustment                  | ✓    | ✓ | Numeric    | ✓ | ✓ |
| P | M | H | C | HUE (Cyan) adjustment                   | ✓    | ✓ | Numeric    | ✓ | ✓ |
| P | M | H | B | HUE (Blue) adjustment                   | ✓    | ✓ | Numeric    | ✓ | ✓ |
| P | M | H | M | HUE (Magenta) adjustment                | ✓    | ✓ | Numeric    | ✓ | ✓ |
| P | M | S | R | SATURATION (Red) adjustment             | ✓    | ✓ | Numeric    | ✓ | ✓ |
| P | M | S | Y | SATURATION (Yellow) adjustment          | ✓    | ✓ | Numeric    | ✓ | ✓ |
| P | M | S | G | SATURATION (Green) adjustment           | ✓    | ✓ | Numeric    | ✓ | ✓ |
| P | M | S | C | SATURATION (Cyan) adjustment            | ✓    | ✓ | Numeric    | ✓ | ✓ |
| P | M | S | B | SATURATION (Blue) adjustment            | ✓    | ✓ | Numeric    | ✓ | ✓ |
| P | M | S | M | SATURATION (Magenta) adjustment         | ✓    | ✓ | Numeric    | ✓ | ✓ |
| P | M | L | R | BRIGHTNESS (Red) adjustment             | ✓    | ✓ | Numeric    | ✓ | ✓ |
| P | M | L | Y | BRIGHTNESS (Yellow) adjustment          | ✓    | ✓ | Numeric    | ✓ | ✓ |
| P | M | L | G | BRIGHTNESS (Green) adjustment           | ✓    | ✓ | Numeric    | ✓ | ✓ |
| P | M | L | C | BRIGHTNESS (Cyan) adjustment            | ✓    | ✓ | Numeric    | ✓ | ✓ |
| P | M | L | B | BRIGHTNESS (Blue) adjustment            | ✓    | ✓ | Numeric    | ✓ | ✓ |
| P | M | L | M | BRIGHTNESS (Magenta) adjustment         | ✓    | ✓ | Numeric    | ✓ | ✓ |
| P | M | L | L | Low Latency                             | ✓    | ✓ | Special    | ✓ | ✓ |
| P | M | C | M | Clear Motion Drive                      | ✓    | ✓ | Special    | ✓ | ✓ |
| P | M | M | E | Motion Enhance                          | ✓    | ✓ | Special    | ✓ | ✓ |
| P | M | L | A | Lens Aperture                           | ✓    | ✓ | Numeric    | ✓ | ✓ |
| P | M | L | P | LD Power (*5)                           | ✓    | ✓ | Special    | ✓ | ✓ |
| P | M | U | S | 8K e-shift                              | ✓    | ✓ | Special    | ✓ | ✓ |
| P | M | G | M | Graphic Mode                            | ✓    | ✓ | Special    | ✓ | ✓ |
| P | M | E | N | Enhance                                 | ✓    | ✓ | Numeric    | ✓ | ✓ |
| P | M | S | T | Smoother                                | ✓    | ✓ | Special    | ✓ | ✓ |
| P | M | U | 1 | Name Edit of Picture Mode User1         | ✓    | ✓ | Special 10 | ✓ | ✓ |
| P | M | U | 2 | Name Edit of Picture Mode User2         | ✓    | ✓ | Special 10 | ✓ | ✓ |
| P | M | U | 3 | Name Edit of Picture Mode User3         | ✓    | ✓ | Special 10 | ✓ | ✓ |
| P | M | U | 4 | Name Edit of Picture Mode User4         | ✓    | ✓ | Special 10 | ✓ | ✓ |
| P | M | U | 5 | Name Edit of Picture Mode User5         | ✓    | ✓ | Special 10 | ✓ | ✓ |
| P | M | U | 6 | Name Edit of Picture Mode User6         | ✓    | ✓ | Special 10 | ✓ | ✓ |
| P | M | C | I | Information of Calibrator               | ✓    | ✓ | Special 3  | ✓ | ✓ |
| P | M | T | L | Mapping Level                           | ✓    | ✓ | Numeric    | ✓ | ✓ |
| P | M | H | L | HDR Level                               | ✓    | ✓ | Special    | ✓ | ✓ |
| P | M | H | P | HDR Processing                          | ✓    | ✓ | Special    | ✓ | ✓ |
| P | M | C | T | Content Type                            | ✓    | ✓ | Special    | ✓ | ✓ |
| P | M | N | M | Theater Optimizer                       | ✓    | ✓ | Special    | ✓ | ✓ |
| P | M | N | L | Theater Optimizer Level                 | ✓    | ✓ | Special    | ✓ | ✓ |
| P | M | N | P | Theater Optimizer Processing            | ✓    | ✓ | Special    | ✓ | ✓ |
| P | M | A | T | Auto transition value for Content Type  | -    | ✓ | Special    | ✓ | ✓ |
| P | M | A | F | Auto transition value for Color Profile | -    | ✓ | Special    | ✓ | ✓ |
| P | M | B | N | Block Noise Reduction                   | ✓    | ✓ | Numeric    | ✓ | ✓ |
| P | M | Q | N | Mosquito Noise Reduction                | ✓    | ✓ | Numeric    | ✓ | ✓ |
| P | M | C | V | LD Current Value                        | ✓    | ✓ | Numeric    | ✓ | ✓ |
| P | M | B | K | Deep Black                              | ✓    | ✓ | Special    | ✓ | ✓ |
| P | M | B | E | Highlight Color                         | ✓    | ✓ | Special    | ✓ | ✓ |
| I | S | I | L | HDMI Input Level switch                 | ✓    | ✓ | Special    | ✓ | ✓ |
| I | S | H | S | HDMI Color Space switch                 | ✓    | ✓ | Special    | ✓ | ✓ |
| I | S | 3 | D | HDMI 2D/3D switch                       | ✓    | ✓ | Special    | ✓ | ✓ |
| I | S | 3 | P | HDMI 3D Phase adjustment                | ✓    | ✓ | Numeric    | ✓ | ✓ |
| I | S | A | S | Aspect switch                           | ✓    | ✓ | Special    | ✓ | ✓ |
| I | S | M | A | Mask switch                             | ✓    | ✓ | Special    | ✓ | ✓ |
| I | S | M | L | Mask (Left) adjustment                  | (*2) | ✓ | Numeric    | ✓ | ✓ |
| I | S | M | R | Mask (Right) adjustment                 | (*2) | ✓ | Numeric    | ✓ | ✓ |
| I | S | M | T | Mask (Top) adjustment                   | (*2) | ✓ | Numeric    | ✓ | ✓ |
| I | S | M | B | Mask (Bottom) adjustment                | (*2) | ✓ | Numeric    | ✓ | ✓ |
| I | S | L | V | Parallax of 3D conversion adjustment    | ✓    | ✓ | Numeric    | ✓ | ✓ |
| I | S | C | A | Crosstalk Cancel (White) adjustment     | ✓    | ✓ | Numeric    | ✓ | ✓ |
| I | S | H | D | HDR10 Picture Mode                      | ✓    | ✓ | Special    | ✓ | ✓ |
| I | S | H | L | HLG Picture Mode                        | ✓    | ✓ | Special    | ✓ | ✓ |
| I | S | S | 2 | SDR(2D) Picture Mode                    | ✓    | ✓ | Special    | ✓ | ✓ |
| I | S | S | 3 | SDR(3D) Picture Mode                    | ✓    | ✓ | Special    | ✓ | ✓ |
| I | S | F | S | FILMMAKER MODE Setting                  | ✓    | ✓ | Special    | ✓ | ✓ |

|     |     |     |     |                                           |   |   |            |   |   |
|-----|-----|-----|-----|-------------------------------------------|---|---|------------|---|---|
| 'I' | 'S' | 'A' | 'L' | ALLM (Auto Low Latency Mode)              | ✓ | ✓ | Special    | ✓ | ✓ |
| 'I' | 'N' | 'F' | 'N' | Focus Near adjustment (*3)                | ✓ | ✓ | Special    | ✓ | ✓ |
| 'I' | 'N' | 'F' | 'F' | Focus Far adjustment (*3)                 | ✓ | ✓ | Special    | ✓ | ✓ |
| 'I' | 'N' | 'Z' | 'T' | Zoom Tele adjustment (*3)                 | ✓ | ✓ | Special    | ✓ | ✓ |
| 'I' | 'N' | 'Z' | 'W' | Zoom Wide adjustment (*3)                 | ✓ | ✓ | Special    | ✓ | ✓ |
| 'I' | 'N' | 'S' | 'L' | Shift Left adjustment (*3)                | ✓ | ✓ | Special    | ✓ | ✓ |
| 'I' | 'N' | 'S' | 'R' | Shift Right adjustment (*3)               | ✓ | ✓ | Special    | ✓ | ✓ |
| 'I' | 'N' | 'S' | 'U' | Shift Up adjustment (*3)                  | ✓ | ✓ | Special    | ✓ | ✓ |
| 'I' | 'N' | 'S' | 'D' | Shift Down adjustment (*3)                | ✓ | ✓ | Special    | ✓ | ✓ |
| 'I' | 'N' | 'I' | 'P' | Image Pattern switch                      | ✓ | ✓ | Special    | ✓ | ✓ |
| 'I' | 'N' | 'L' | 'L' | Lens Lock switch                          | ✓ | ✓ | Special    | ✓ | ✓ |
| 'I' | 'N' | 'X' | 'R' | Pixel Adjust (Horizontal Red) adjustment  | ✓ | ✓ | Numeric    | ✓ | ✓ |
| 'I' | 'N' | 'X' | 'B' | Pixel Adjust (Horizontal Blue) adjustment | ✓ | ✓ | Numeric    | ✓ | ✓ |
| 'I' | 'N' | 'Y' | 'R' | Pixel Adjust (Vertical Red) adjustment    | ✓ | ✓ | Numeric    | ✓ | ✓ |
| 'I' | 'N' | 'Y' | 'B' | Pixel Adjust (Vertical Blue) adjustment   | ✓ | ✓ | Numeric    | ✓ | ✓ |
| 'I' | 'N' | 'P' | '2' | Image Pattern for Pixel Adjust switch     | ✓ | ✓ | Special    | ✓ | ✓ |
| 'I' | 'N' | 'I' | 'S' | Installation Style switch                 | ✓ | ✓ | Special    | ✓ | ✓ |
| 'I' | 'N' | 'K' | 'V' | Keystone (Vertical) adjustment            | ✓ | ✓ | Numeric    | ✓ | ✓ |
| 'I' | 'N' | 'V' | 'S' | Anamorphic switch                         | ✓ | ✓ | Special    | ✓ | ✓ |
| 'I' | 'N' | 'S' | 'A' | Screen Adjust Data                        | ✓ | ✓ | Numeric    | ✓ | ✓ |
| 'I' | 'N' | 'S' | 'C' | Screen Adjust switch                      | ✓ | ✓ | Special    | ✓ | ✓ |
| 'I' | 'N' | 'P' | 'A' | Panel Alignment switch                    | ✓ | ✓ | Special    | ✓ | ✓ |
| 'I' | 'N' | 'M' | 'L' | Installation Mode Load                    | ✓ | - | Special    | ✓ | ✓ |
| 'I' | 'N' | 'M' | '1' | Name Edit of Installation Mode 1          | ✓ | ✓ | Special 10 | ✓ | ✓ |
| 'I' | 'N' | 'M' | '2' | Name Edit of Installation Mode 2          | ✓ | ✓ | Special 10 | ✓ | ✓ |
| 'I' | 'N' | 'M' | '3' | Name Edit of Installation Mode 3          | ✓ | ✓ | Special 10 | ✓ | ✓ |
| 'I' | 'N' | 'M' | '4' | Name Edit of Installation Mode 4          | ✓ | ✓ | Special 10 | ✓ | ✓ |
| 'I' | 'N' | 'M' | '5' | Name Edit of Installation Mode 5          | ✓ | ✓ | Special 10 | ✓ | ✓ |
| 'I' | 'N' | 'M' | '6' | Name Edit of Installation Mode 6          | ✓ | ✓ | Special 10 | ✓ | ✓ |
| 'I' | 'N' | 'M' | '7' | Name Edit of Installation Mode 7          | ✓ | ✓ | Special 10 | ✓ | ✓ |
| 'I' | 'N' | 'M' | '8' | Name Edit of Installation Mode 8          | ✓ | ✓ | Special 10 | ✓ | ✓ |
| 'I' | 'N' | 'M' | '9' | Name Edit of Installation Mode 9          | ✓ | ✓ | Special 10 | ✓ | ✓ |
| 'I' | 'N' | 'M' | 'A' | Name Edit of Installation Mode 10         | ✓ | ✓ | Special 10 | ✓ | ✓ |
| 'I' | 'N' | 'I' | 'N' | Focus Near adjustment (1 shot) (*3)       | ✓ | ✓ | None       | ✓ | ✓ |
| 'I' | 'N' | 'I' | 'F' | Focus Far adjustment (1 shot) (*3)        | ✓ | ✓ | None       | ✓ | ✓ |
| 'I' | 'N' | 'I' | 'T' | Zoom Tele adjustment (1 shot) (*3)        | ✓ | ✓ | None       | ✓ | ✓ |
| 'I' | 'N' | 'I' | 'W' | Zoom Wide adjustment (1 shot) (*3)        | ✓ | ✓ | None       | ✓ | ✓ |
| 'I' | 'N' | 'I' | 'L' | Shift Left adjustment (1 shot) (*3)       | ✓ | ✓ | None       | ✓ | ✓ |
| 'I' | 'N' | 'I' | 'R' | Shift Right adjustment (1 shot) (*3)      | ✓ | ✓ | None       | ✓ | ✓ |
| 'I' | 'N' | 'I' | 'U' | Shift Up adjustment (1 shot) (*3)         | ✓ | ✓ | None       | ✓ | ✓ |
| 'I' | 'N' | 'I' | 'D' | Shift Down adjustment (1 shot) (*3)       | ✓ | ✓ | None       | ✓ | ✓ |
| 'I' | 'N' | 'H' | 'A' | High Altitude mode switch                 | ✓ | ✓ | Special    | ✓ | ✓ |
| 'I' | 'N' | 'N' | 'S' | Screen Size                               | ✓ | ✓ | Special 4  | ✓ | ✓ |
| 'I' | 'N' | 'N' | 'G' | Screen Gain                               | ✓ | ✓ | Special 4  | ✓ | ✓ |
| 'I' | 'N' | 'N' | 'A' | Screen Aspect                             | ✓ | ✓ | Special    | ✓ | ✓ |
| 'D' | 'S' | 'B' | 'C' | Back Color switch                         | ✓ | ✓ | Special    | ✓ | ✓ |
| 'D' | 'S' | 'M' | 'P' | Menu Position switch                      | ✓ | ✓ | Special    | ✓ | ✓ |
| 'D' | 'S' | 'S' | 'D' | Source Display switch                     | ✓ | ✓ | Special    | ✓ | ✓ |
| 'D' | 'S' | 'L' | 'O' | Logo switch                               | ✓ | ✓ | Special    | ✓ | ✓ |
| 'D' | 'S' | 'L' | 'A' | Language switch                           | ✓ | ✓ | Special    | ✓ | ✓ |
| 'D' | 'S' | 'D' | 'M' | OSD Dark Mode                             | ✓ | ✓ | Special    | ✓ | ✓ |
| 'D' | 'S' | 'B' | 'L' | OSD Brightness Sync                       | ✓ | ✓ | Special    | ✓ | ✓ |
| 'F' | 'U' | 'T' | 'R' | Trigger switch                            | ✓ | ✓ | Special    | ✓ | ✓ |
| 'F' | 'U' | 'O' | 'T' | Off Timer switch                          | ✓ | ✓ | Special    | ✓ | ✓ |
| 'F' | 'U' | 'E' | 'M' | Eco Mode switch                           | ✓ | ✓ | Special    | ✓ | ✓ |
| 'F' | 'U' | 'H' | 'E' | Hide Eco switch                           | ✓ | ✓ | Special    | ✓ | ✓ |
| 'F' | 'U' | 'C' | 'F' | Control4                                  | ✓ | ✓ | Special    | ✓ | ✓ |
| 'I' | 'F' | 'I' | 'N' | Input display                             | - | ✓ | Special    | ✓ | ✓ |
| 'I' | 'F' | 'I' | 'S' | Source display                            | - | ✓ | Special 4  | ✓ | ✓ |
| 'I' | 'F' | 'R' | 'H' | Horizontal Resolution display             | - | ✓ | Numeric    | ✓ | ✓ |
| 'I' | 'F' | 'R' | 'V' | Vertical Resolution display               | - | ✓ | Numeric    | ✓ | ✓ |
| 'I' | 'F' | 'F' | 'H' | Horizontal Frequency display (*4)         | - | ✓ | Numeric    | ✓ | ✓ |
| 'I' | 'F' | 'F' | 'V' | Vertical Frequency display (*4)           | - | ✓ | Numeric    | ✓ | ✓ |
| 'I' | 'F' | 'D' | 'C' | Color Depth display                       | - | ✓ | Special    | ✓ | ✓ |
| 'I' | 'F' | 'X' | 'V' | Color space display                       | - | ✓ | Special    | ✓ | ✓ |
| 'I' | 'F' | 'C' | 'M' | Colorimetry                               | - | ✓ | Special    | ✓ | ✓ |
| 'I' | 'F' | 'H' | 'R' | HDR                                       | - | ✓ | Special    | ✓ | ✓ |

|     |     |     |     |                      |   |   |            |   |   |
|-----|-----|-----|-----|----------------------|---|---|------------|---|---|
| 'I' | 'F' | 'L' | 'T' | Light Source Time    | - | ✓ | Numeric    | ✓ | ✓ |
| 'I' | 'F' | 'S' | 'V' | Soft Version Display | - | ✓ | Special 14 | ✓ | ✓ |
| 'I' | 'F' | 'M' | 'C' | Max CLL              | - | ✓ | Numeric    | ✓ | ✓ |
| 'I' | 'F' | 'M' | 'F' | Max FALL             | - | ✓ | Numeric    | ✓ | ✓ |
| 'I' | 'F' | 'D' | 'S' | DSC Mode             | - | ✓ | Special    | ✓ | ✓ |
| 'I' | 'F' | 'L' | 'R' | Link Rate            | - | ✓ | Special    | ✓ | ✓ |

(\*1) Only the parameter that follows Picture Mode is effective. (Refer to the table of Picture Mode vs. Color Profile that described in "Color Profile" section of Functional Spec.)

[Example] Picture Mode = Natural Video(0x38) -> Accepted, Anime1(0x36) -> Rejected

Picture Mode = Film Film1(0x31) -> Accepted, Standard(0x33) -> Rejected

If the corresponded parameter is only one, PJ ignores setting command.

(\*2) Setting operations of Mask Left / Right / Top / Bottom [ISML, ISMR, ISMT, ISMB] command are only effective when Mask Setting [ISMA] is set to "On".

(\*3) Because of electrical limitation, only one motor can be driven at the same time. (Same as 1 shot mode)

If the projector receives a motor drive request when other motor is driving, the projector rejects its request.

When a driving motor reaches its limit, the projector stops the motor automatically.

(\*4) Parameter is equal to the result in which 100 is multiplied with the actual value.

[Example] When Horz. Frequency is 63.98 kHz : Parameter =  $63.98 * 100 = 6398 = 0x18FE$

(\*5) LD Power sets the current value corresponding to the conventional parameters (Low/Mid/High).

## Parameter2

Parameter2 data depends on Sub command.

Data length is as follow depending on Sub command.

| Data type  | Data length | Note                                         |
|------------|-------------|----------------------------------------------|
| Numeric    | 4 bytes     | ASCII character                              |
| Special    | 1 - 4 byte  | ASCII character                              |
| Special 2  | 512 bytes   | Binary data(for Gamma)                       |
| Special 3  | 18 bytes    | ASCII character (Information for Calibrator) |
| Special 4  | 2 bytes     | ASCII character                              |
| Special 10 | 10 bytes    | ASCII character                              |
| Special 14 | 6 bytes     | ASCII character                              |

- ◆ When the projector is not powered on and it receives the command, the projector ignores the command and does not respond.

## 4.10.1 Special data

**Picture mode**

Table 4-19 PICTURE MODE CMD DATA

| HEX       | ASCII | Operation                     |
|-----------|-------|-------------------------------|
| 0x30 0x30 | 00    | Film                          |
| 0x30 0x31 | 01    | Cinema                        |
| 0x30 0x33 | 03    | Natural                       |
| 0x30 0x34 | 04    | HDR10                         |
| 0x30 0x42 | 0B    | Frame Adapt HDR               |
| 0x30 0x43 | 0C    | User1                         |
| 0x30 0x44 | 0D    | User2                         |
| 0x30 0x45 | 0E    | User3                         |
| 0x30 0x46 | 0F    | User4                         |
| 0x31 0x30 | 10    | User5                         |
| 0x31 0x31 | 11    | User6                         |
| 0x31 0x34 | 14    | HLG                           |
| 0x31 0x35 | 15    | HDR10+                        |
| 0x31 0x37 | 17    | FILMMAKER MODE                |
| 0x31 0x38 | 18    | Frame Adapt HDR 2             |
| 0x31 0x39 | 19    | Frame Adapt HDR 3 / HDR Vivid |
| 0x31 0x42 | 1B    | Vivid                         |
| 0x31 0x43 | 1C    | Natural (LL)                  |
| 0x31 0x44 | 1D    | HDR10 (LL)                    |
| 0x31 0x45 | 1E    | HLG (LL)                      |

**Color Profile**

Table 4-20 COLOR PROFILE CMD DATA

| HEX       | ASCII | Operation        |
|-----------|-------|------------------|
| 0x30 0x30 | 00    | Off (Normal)     |
| 0x30 0x31 | 01    | Film1            |
| 0x30 0x32 | 02    | Film2            |
| 0x30 0x33 | 03    | BT.709           |
| 0x30 0x34 | 04    | Cinema           |
| 0x30 0x36 | 06    | Anime            |
| 0x30 0x38 | 08    | Video            |
| 0x30 0x42 | 0B    | BT.2020 (Wide)   |
| 0x30 0x45 | 0E    | Custom1          |
| 0x30 0x46 | 0F    | Custom2          |
| 0x31 0x30 | 10    | Custom3          |
| 0x31 0x31 | 11    | Custom4          |
| 0x32 0x31 | 21    | DCI              |
| 0x32 0x33 | 23    | Vivid            |
| 0x32 0x34 | 24    | BT.2020 (Normal) |
| 0x32 0x35 | 25    | Off (Wide)       |
| 0x32 0x36 | 26    | Auto             |

**Color Temp. Table Data**

Table 4-21 COLOR TEMP TABLE CMD DATA

| HEX       | ASCII | Operation   |
|-----------|-------|-------------|
| 0x30 0x30 | 00    | 5500K       |
| 0x30 0x32 | 02    | 6500K       |
| 0x30 0x34 | 04    | 7500K       |
| 0x30 0x38 | 08    | 9300K       |
| 0x30 0x39 | 09    | High Bright |
| 0x30 0x41 | 0A    | Custom1     |
| 0x30 0x42 | 0B    | Custom2     |
| 0x30 0x43 | 0C    | HDR10       |
| 0x30 0x44 | 0D    | Xenon1      |
| 0x30 0x45 | 0E    | Xenon2      |
| 0x30 0x34 | 14    | HLG         |

|           |     |        |
|-----------|-----|--------|
| 0x30 0x35 | 1 5 | HDR10+ |
|-----------|-----|--------|

## Color Temp. Correction Data

Table 4-22 COLOR TEMP CORRECTION CMD DATA

| HEX       | ASCII | Operation   |
|-----------|-------|-------------|
| 0x30 0x30 | '0 0' | 5500K       |
| 0x30 0x32 | '0 2' | 6500K       |
| 0x30 0x34 | '0 4' | 7500K       |
| 0x30 0x38 | '0 8' | 9300K       |
| 0x30 0x39 | '0 9' | High Bright |
| 0x30 0x44 | '0 D' | Xenon1      |
| 0x30 0x45 | '0 E' | Xenon2      |

## Gamma Table Data

Table 4-23 GAMMA CMD DATA

| HEX       | ASCII | Operation |
|-----------|-------|-----------|
| 0x30 0x30 | '0 0' | 2.2       |
| 0x30 0x31 | '0 1' | Cinema1   |
| 0x30 0x32 | '0 2' | Cinema2   |
| 0x30 0x34 | '0 4' | Custom1   |
| 0x30 0x35 | '0 5' | Custom2   |
| 0x30 0x36 | '0 6' | Custom3   |
| 0x30 0x37 | '0 7' | HDR(HLG)  |
| 0x30 0x38 | '0 8' | 2.4       |
| 0x30 0x39 | '0 9' | 2.6       |
| 0x30 0x41 | '0 A' | Film1     |
| 0x30 0x42 | '0 B' | Film2     |
| 0x30 0x43 | '0 C' | HDR(PQ)   |
| 0x31 0x35 | '1 5' | HDR(Auto) |
| 0x31 0x36 | '1 6' | Vivid     |

## Gamma Correction Data

Table 4-24 GAMMA CORRECTION CMD DATA

| HEX       | ASCII | Operation |
|-----------|-------|-----------|
| 0x30 0x31 | '0 1' | Cinema1   |
| 0x30 0x32 | '0 2' | Cinema2   |
| 0x30 0x34 | '0 4' | Import    |
| 0x30 0x35 | '0 5' | 1.8       |
| 0x30 0x36 | '0 6' | 1.9       |
| 0x30 0x37 | '0 7' | 2.0       |
| 0x30 0x38 | '0 8' | 2.1       |
| 0x30 0x39 | '0 9' | 2.2       |
| 0x30 0x41 | '0 A' | 2.3       |
| 0x30 0x42 | '0 B' | 2.4       |
| 0x30 0x43 | '0 C' | 2.5       |
| 0x30 0x44 | '0 D' | 2.6       |
| 0x30 0x45 | '0 E' | Film1     |
| 0x30 0x46 | '0 F' | Film2     |
| 0x31 0x34 | '1 4' | HDR(HLG)  |
| 0x31 0x35 | '1 5' | HDR(PQ)   |
| 0x31 0x37 | '1 7' | Vivid     |



## Dynamic CTRL

Table 4-25 DYNAMIC CTRL CMD DATA

| HEX  | ASCII | Operation |
|------|-------|-----------|
| 0x30 | 0     | Off       |
| 0x31 | 1     | Low       |
| 0x32 | 2     | High      |
| 0x33 | 3     | Balanced  |

## Color Management Data

Table 4-26 COLOR MANAGEMENT CMD DATA

| HEX  | ASCII | Operation |
|------|-------|-----------|
| 0x30 | '0'   | Off       |
| 0x31 | '1'   | On        |

## Low Latency Data

Table 4-27 LOW LATENCY CMD DATA

| HEX  | ASCII | Operation |
|------|-------|-----------|
| 0x30 | '0'   | Off       |
| 0x31 | '1'   | On        |

## Clear Motion Drive Data

Table 4-28 CLEAR MOTION DRIVE CMD DATA

| HEX  | ASCII | Operation        |
|------|-------|------------------|
| 0x30 | '0'   | Off              |
| 0x33 | '3'   | Low              |
| 0x34 | '4'   | High             |
| 0x35 | '5'   | Inverse Telecine |

## Motion Enhance Data

Table 4-29 MOTION ENHANCE CMD DATA

| HEX  | ASCII | Operation |
|------|-------|-----------|
| 0x30 | '0'   | Off       |
| 0x31 | '1'   | Low       |
| 0x32 | '2'   | High      |

## LD Power Data

Table 4-30 LD POWER CMD DATA

| HEX  | ASCII | Operation |
|------|-------|-----------|
| 0x30 | '0'   | 109       |
| 0x31 | '1'   | 219       |
| 0x32 | '2'   | 160       |

## 8K e-shift Data

Table 4-31 8K E-SHIFT CMD DATA

| HEX  | ASCII | Operation |
|------|-------|-----------|
| 0x30 | '0'   | Off       |
| 0x31 | '1'   | On        |

## Graphic Mode Data

Table 4-32 GRAPHIC MODE CMD DATA

| HEX  | ASCII | Operation |
|------|-------|-----------|
| 0x30 | '0'   | Standard  |
| 0x31 | '1'   | High-res  |
| 0x32 | '2'   | High-res2 |
| 0x46 | 'F'   | Off       |

\*"Off" is valid only for reference when Picture Mode is "FILMMAKER MODE". (Operation is invalid)

## Smoother Data

Table 4-33 SMOOTHER CMD DATA

| HEX                 | ASCII     | Operation |
|---------------------|-----------|-----------|
| 0x30 0x30 0x30 0x30 | '0 0 0 0' | On        |
| 0x30 0x30 0x30 0x31 | '0 0 0 1' | Off       |

## HDR Level Data

Table 4-34 HDR Level CMD DATA

| HEX  | ASCII | Operation                                                 |                     |
|------|-------|-----------------------------------------------------------|---------------------|
|      |       | Frame Adapt HDR<br>Frame Adapt HDR 2<br>Frame Adapt HDR 3 | HDR10+<br>HDR V'vid |
| 0x30 | '0'   | Auto(Normal)                                              | -                   |
| 0x31 | '1'   | -2                                                        | -                   |
| 0x32 | '2'   | -1                                                        | Low                 |
| 0x33 | '3'   | 0                                                         | Mid                 |
| 0x34 | '4'   | 1                                                         | High                |
| 0x35 | '5'   | 2                                                         | -                   |
| 0x36 | '6'   | Auto(Wide)                                                | -                   |

## HDR Processing Data

Table 4-35 HDR PROCESSING CMD DATA

| HEX  | ASCII | Operation      |
|------|-------|----------------|
| 0x30 | '0'   | HDR10+         |
| 0x31 | '1'   | Static         |
| 0x32 | '2'   | Frame by Frame |
| 0x33 | '3'   | Scene by Scene |

\*"HDR10+" is valid only for reference when Picture Mode is "HDR10+". (Operation is invalid)

## Content Type Data

Table 4-36 CONTNT TYPE CMD DATA

| HEX  | ASCII | Operation |
|------|-------|-----------|
| 0x30 | '0'   | Auto      |
| 0x31 | '1'   | SDR       |
| 0x32 | '2'   | HDR10+    |
| 0x33 | '3'   | HDR10     |
| 0x34 | '4'   | HLG       |

## Theater Optimizer Data

Table 4-37 THEATER OPTIMIZER CMD DATA

| HEX  | ASCII | Operation |
|------|-------|-----------|
| 0x30 | '0'   | Off       |
| 0x31 | '1'   | On        |

## Theater Optimizer Level Data

Table 4-38 THEATER OPTIMIZER LEVEL CMD DATA

| HEX  | ASCII | Operation    |
|------|-------|--------------|
| 0x30 | '0'   | Auto(Normal) |
| 0x31 | '1'   | -2           |
| 0x32 | '2'   | -1           |
| 0x33 | '3'   | 0            |
| 0x34 | '4'   | 1            |
| 0x35 | '5'   | 2            |
| 0x36 | '6'   | Auto(Wide)   |

## Theater Optimizer Processing

Table 4-39 THEATER OPTIMIZER PROCESSING

| HEX  | ASCII | Operation        |
|------|-------|------------------|
| 0x30 | '0'   | -                |
| 0x31 | '1'   | Processing Start |

## Auto transition value for Content Type Data

Table 4-40 AUTO TRANSITION VALUE FOR CONTENT TYPE DATA

| HEX  | ASCII | Operation |
|------|-------|-----------|
| 0x31 | '1'   | SDR       |
| 0x32 | '2'   | HDR10+    |
| 0x33 | '3'   | HDR10     |
| 0x34 | '4'   | HLG       |

## Auto transition value for Color Profile Data

Table 4-41 AUTO TRANSITION VALUE FOR COLOR PROFILE DATA

| HEX       | ASCII | Operation       |
|-----------|-------|-----------------|
| 0x30 0x30 | 00    | Off(Normal)     |
| 0x30 0x31 | 01    | Film1           |
| 0x30 0x32 | 02    | Film2           |
| 0x30 0x33 | 03    | BT.709          |
| 0x30 0x34 | 04    | Cinema          |
| 0x30 0x36 | 06    | Anime           |
| 0x30 0x38 | 08    | Video           |
| 0x30 0x41 | 0A    | HDR             |
| 0x30 0x42 | 0B    | BT.2020(Wide)   |
| 0x30 0x45 | 0E    | Custom1         |
| 0x30 0x46 | 0F    | Custom2         |
| 0x31 0x30 | 10    | Custom3         |
| 0x31 0x31 | 11    | Custom4         |
| 0x32 0x31 | 21    | DCI             |
| 0x32 0x33 | 23    | Vivid           |
| 0x32 0x34 | 24    | BT.2020(Normal) |
| 0x32 0x35 | 25    | Off(Wide)       |

## Deep Black Data

Table 4-42 DEEP BLACK CMD DATA

| HEX  | ASCII | Operation |
|------|-------|-----------|
| 0x30 | '0'   | Off       |
| 0x31 | '1'   | On        |

## Highlight Color Data

Table 4-43 HIGHLIGHT COLOR CMD DATA

| HEX  | ASCII | Operation |
|------|-------|-----------|
| 0x30 | '0'   | Low       |
| 0x31 | '1'   | Mid       |
| 0x32 | '2'   | High      |

## HDMI Input Level Data

Table 4-44 HDMI INPUT LEVEL CMD DATA

| HEX  | ASCII | Operation            |
|------|-------|----------------------|
| 0x30 | '0'   | Standard (16-235)    |
| 0x31 | '1'   | Enhanced (0-255)     |
| 0x32 | '2'   | Super White (16-255) |
| 0x33 | '3'   | Auto                 |

## HDMI Color Space Data

Table 4-45 HDMI COLOR SPACE CMD DATA

| HEX  | ASCII | Operation    |
|------|-------|--------------|
| 0x30 | '0'   | Auto         |
| 0x31 | '1'   | YCbCr(4:4:4) |
| 0x32 | '2'   | YCbCr(4:2:2) |
| 0x33 | '3'   | RGB          |

## HDMI 2D/3D Data

Table 4-46 HDMI 2D/3D CMD DATA

| HEX  | ASCII | Operation      |
|------|-------|----------------|
| 0x30 | '0'   | 2D             |
| 0x31 | '1'   | Auto           |
| 0x33 | '3'   | Side By Side   |
| 0x34 | '4'   | Top and Bottom |

## Aspect Data

Table 4-47 ASPECT CMD DATA

| HEX  | ASCII | Operation |
|------|-------|-----------|
| 0x32 | '2'   | Zoom      |
| 0x33 | '3'   | AUTO      |
| 0x34 | '4'   | Native    |

## Mask Data

Table 4-48 MASK CMD DATA

| HEX  | ASCII | Operation |
|------|-------|-----------|
| 0x31 | '1'   | On        |
| 0x32 | '2'   | Off       |

## HDR10 Picture Mode Data

Table 4-49 HDR10 PICTURE MODE CMD DATA

| HEX  | ASCII | Operation |
|------|-------|-----------|
| 0x31 | '1'   | HDR10     |

|      |     |                               |
|------|-----|-------------------------------|
| 0x35 | '5' | USER4                         |
| 0x36 | '6' | USER5                         |
| 0x37 | '7' | USER6                         |
| 0x38 | '8' | Frame Adapt HDR               |
| 0x43 | 'C' | Frame Adapt HDR 2             |
| 0x44 | 'D' | Frame Adapt HDR 3 / HDR Vivid |
| 0x45 | 'E' | HDR10 (LL)                    |
| 0x46 | 'F' | Last Setting                  |

## HLG Picture Mode Data

Table 4-50 HLG PICTURE MODE CMD DATA

| HEX  | ASCII | Operation    |
|------|-------|--------------|
| 0x31 | '1'   | HLG          |
| 0x35 | '5'   | USER4        |
| 0x36 | '6'   | USER5        |
| 0x37 | '7'   | USER6        |
| 0x39 | '9'   | HLG (LL)     |
| 0x46 | 'F'   | Last Setting |

## SDR(2D) Picture Mode Data

Table 4-51 SDR(2D) PICTURE MODE CMD DATA

| HEX  | ASCII | Operation    |
|------|-------|--------------|
| 0x31 | '1'   | Natural      |
| 0x32 | '2'   | USER1        |
| 0x33 | '3'   | USER2        |
| 0x34 | '4'   | USER3        |
| 0x38 | '8'   | Cinema       |
| 0x39 | '9'   | Film         |
| 0x44 | 'D'   | Natural (LL) |
| 0x45 | 'E'   | Vivid        |
| 0x46 | 'F'   | Last Setting |

## SDR(3D) Picture Mode Data

Table 4-52 SDR(3D) PICTURE MODE CMD DATA

| HEX  | ASCII | Operation    |
|------|-------|--------------|
| 0x31 | '1'   | Natural      |
| 0x32 | '2'   | USER1        |
| 0x33 | '3'   | USER2        |
| 0x34 | '4'   | USER3        |
| 0x38 | '8'   | Cinema       |
| 0x39 | '9'   | Film         |
| 0x44 | 'D'   | Natural (LL) |
| 0x45 | 'E'   | Vivid        |
| 0x46 | 'F'   | Last Setting |

## FILMMAKER MODE Setting

Table 4-53 FILMMAKER MODE Setting CMD DATA

| HEX  | ASCII | Operation |
|------|-------|-----------|
| 0x30 | '0'   | Manual    |
| 0x31 | '1'   | Auto      |

## ALLM (Auto Low Latency Mode)

Table 4-54 ALLM CMD DATA

| HEX  | ASCII | Operation |
|------|-------|-----------|
| 0x30 | '0'   | Off       |
| 0x31 | '1'   | On        |

## Lens Control (Focus, Zoom, Shift) Data

Table 4-55 LENS CONTROL (FOCUS / ZOOM / SHIFT) CMD DATA

| HEX  | ASCII | Operation |
|------|-------|-----------|
| 0x30 | '0'   | Stop      |
| 0x31 | '1'   | Start     |

## Lens Image Pattern, Lens Lock, Screen Adjust Data

Table 4-56 Above CMD DATA

| HEX  | ASCII | Operation |
|------|-------|-----------|
| 0x30 | '0'   | Off       |
| 0x31 | '1'   | On        |

## Installation Style Data

Table 4-57 INSTALLATION STYLE CMD DATA

| HEX  | ASCII | Operation         |
|------|-------|-------------------|
| 0x30 | '0'   | Front             |
| 0x31 | '1'   | Ceiling Mount (F) |
| 0x32 | '2'   | Rear              |
| 0x33 | '3'   | Ceiling Mount (R) |

## Anamorphic Data

Table 4-58 ANAMORPHIC CMD DATA

| HEX  | ASCII | Operation |
|------|-------|-----------|
| 0x30 | '0'   | Off       |
| 0x31 | '1'   | A         |
| 0x32 | '2'   | B         |
| 0x33 | '3'   | C         |
| 0x34 | '4'   | D         |

## Screen Adjust Switch Data

Table 4-59 SCREEN ADJUST CMD DATA

| HEX  | ASCII | Operation |
|------|-------|-----------|
| 0x30 | '0'   | Off       |
| 0x31 | '1'   | On        |

## Panel Alignment Switch Data

Table 4-60 PANEL ALIGNMENT CMD DATA

| HEX  | ASCII | Operation |
|------|-------|-----------|
| 0x30 | '0'   | Off       |
| 0x31 | '1'   | On        |

## Installation Mode Load Data

Table 4-61 INSTALLATION MODE LOAD CMD DATA

| HEX  | ASCII | Operation |
|------|-------|-----------|
| 0x30 | '0'   | Memory 1  |
| 0x31 | '1'   | Memory 2  |
| 0x32 | '2'   | Memory 3  |
| 0x33 | '3'   | Memory 4  |

|      |     |           |
|------|-----|-----------|
| 0x34 | '4' | Memory 5  |
| 0x35 | '5' | Memory 6  |
| 0x36 | '6' | Memory 7  |
| 0x37 | '7' | Memory 8  |
| 0x38 | '8' | Memory 9  |
| 0x39 | '9' | Memory 10 |

## High Altitude mode Data

Table 4-62 HIGH ALTITUDE CMD DATA

| HEX  | ASCII | Operation |
|------|-------|-----------|
| 0x30 | '0'   | Off       |
| 0x31 | '1'   | On        |

## Screen Size Data

Table 4-63 SCREEN SIZE DATA

| HEX       | ASCII | Operation | NZ900 etc.. | NZ800 etc.. |
|-----------|-------|-----------|-------------|-------------|
| 0x30 0x36 | 06    | 60 inch   | ✓           | ✓           |
| 0x30 0x37 | 07    | 70 inch   | ✓           | ✓           |
| 0x30 0x38 | 08    | 80 inch   | ✓           | ✓           |
| 0x30 0x39 | 09    | 90 inch   | ✓           | ✓           |
| 0x30 0x41 | 0A    | 100 inch  | ✓           | ✓           |
| 0x30 0x42 | 0B    | 110 inch  | ✓           | ✓           |
| 0x30 0x43 | 0C    | 120 inch  | ✓           | ✓           |
| 0x30 0x44 | 0D    | 130 inch  | ✓           | ✓           |
| 0x30 0x45 | 0E    | 140 inch  | ✓           | ✓           |
| 0x30 0x46 | 0F    | 150 inch  | ✓           | ✓           |
| 0x31 0x30 | 10    | 160 inch  | ✓           | ✓           |
| 0x31 0x31 | 11    | 170 inch  | ✓           | ✓           |
| 0x31 0x32 | 12    | 180 inch  | ✓           | ✓           |
| 0x31 0x33 | 13    | 190 inch  | ✓           | ✓           |
| 0x31 0x34 | 14    | 200 inch  | ✓           | ✓           |
| 0x31 0x35 | 15    | 210 inch  | ✓           | -           |
| 0x31 0x36 | 16    | 220 inch  | ✓           | -           |
| 0x31 0x37 | 17    | 230 inch  | ✓           | -           |
| 0x31 0x38 | 18    | 240 inch  | ✓           | -           |
| 0x31 0x39 | 19    | 250 inch  | ✓           | -           |
| 0x31 0x41 | 1A    | 260 inch  | ✓           | -           |
| 0x31 0x42 | 1B    | 270 inch  | ✓           | -           |
| 0x31 0x43 | 1C    | 280 inch  | ✓           | -           |
| 0x31 0x44 | 1D    | 290 inch  | ✓           | -           |
| 0x31 0x45 | 1E    | 300 inch  | ✓           | -           |

## Screen Gain Data

Table 4-64 SCREEN GAIN DATA

| HEX       | ASCII | Operation |
|-----------|-------|-----------|
| 0x30 0x38 | 08    | 0.8       |
| 0x30 0x39 | 09    | 0.9       |

|           |    |     |
|-----------|----|-----|
| 0x30 0x41 | 0A | 1.0 |
| 0x30 0x42 | 0B | 1.1 |
| 0x30 0x43 | 0C | 1.2 |
| 0x30 0x44 | 0D | 1.3 |
| 0x30 0x45 | 0E | 1.4 |
| 0x30 0x46 | 0F | 1.5 |
| 0x31 0x30 | 10 | 1.6 |
| 0x31 0x31 | 11 | 1.7 |
| 0x31 0x32 | 12 | 1.8 |
| 0x31 0x33 | 13 | 1.9 |
| 0x31 0x34 | 14 | 2.0 |
| 0x31 0x35 | 15 | 2.1 |
| 0x31 0x36 | 16 | 2.2 |
| 0x31 0x37 | 17 | 2.3 |
| 0x31 0x38 | 18 | 2.4 |
| 0x31 0x39 | 19 | 2.5 |
| 0x31 0x41 | 1A | 2.6 |
| 0x31 0x42 | 1B | 2.7 |
| 0x31 0x43 | 1C | 2.8 |
| 0x31 0x44 | 1D | 2.9 |
| 0x31 0x45 | 1E | 3.0 |
| 0x31 0x46 | 1F | 3.1 |
| 0x32 0x30 | 20 | 3.2 |
| 0x32 0x31 | 21 | 3.3 |
| 0x32 0x32 | 22 | 3.4 |
| 0x32 0x33 | 23 | 3.5 |
| 0x32 0x34 | 24 | 3.6 |
| 0x32 0x35 | 25 | 3.7 |
| 0x32 0x36 | 26 | 3.8 |
| 0x32 0x37 | 27 | 3.9 |
| 0x32 0x38 | 28 | 4.0 |
| 0x32 0x39 | 29 | 4.1 |
| 0x32 0x41 | 2A | 4.2 |
| 0x32 0x42 | 2B | 4.3 |
| 0x32 0x43 | 2C | 4.4 |
| 0x32 0x44 | 2D | 4.5 |
| 0x32 0x45 | 2E | 4.6 |
| 0x32 0x46 | 2F | 4.7 |
| 0x33 0x30 | 30 | 4.8 |
| 0x33 0x31 | 31 | 4.9 |
| 0x33 0x32 | 32 | 5.0 |

## Screen Aspect Data

Table 4-65 SCREEN ASPECT CMD DATA

| HEX  | ASCII | Operation    |
|------|-------|--------------|
| 0x30 | '0'   | 16:9         |
| 0x31 | '1'   | 17:9         |
| 0x32 | '2'   | Cinema Scope |

## Back Color Data

Table 4-66 BACK COLOR CMD DATA

| HEX  | ASCII | Operation |
|------|-------|-----------|
| 0x30 | '0'   | Blue      |
| 0x31 | '1'   | Black     |

## Menu Position Data

Table 4-67 MENU POSITION CMD DATA

| HEX  | ASCII | Operation |
|------|-------|-----------|
| 0x30 | '0'   | Left-Top  |



|      |     |              |
|------|-----|--------------|
| 0x31 | '1' | Right-Top    |
| 0x32 | '2' | Center       |
| 0x33 | '3' | Left-Bottom  |
| 0x34 | '4' | Right-Bottom |
| 0x35 | '5' | Left         |
| 0x36 | '6' | Right        |

## Source Display, Logo Data

Table 4-68 SOURCE DISPLAY, LOGO CMD DATA

| HEX  | ASCII | Operation |
|------|-------|-----------|
| 0x30 | '0'   | Off       |
| 0x31 | '1'   | On        |

## Language Data

Table 4-69 LANGUAGE CMD DATA

| HEX  | ASCII | Operation                     |
|------|-------|-------------------------------|
| 0x30 | '0'   | Japanese                      |
| 0x31 | '1'   | English                       |
| 0x32 | '2'   | German                        |
| 0x33 | '3'   | Spanish                       |
| 0x34 | '4'   | Italian                       |
| 0x35 | '5'   | French                        |
| 0x36 | '6'   | Portuguese                    |
| 0x37 | '7'   | Dutch                         |
| 0x38 | '8'   | Polski                        |
| 0x39 | '9'   | Norwegian                     |
| 0x42 | 'B'   | Chinese (Simplified Chinese)  |
| 0x43 | 'C'   | Chinese (Traditional Chinese) |

## OSD Dark Mode Data

Table 4-70 OSD DARK MODE CMD DATA

| HEX  | ASCII | Operation |
|------|-------|-----------|
| 0x30 | '0'   | Off       |
| 0x31 | '1'   | On        |

## OSD Brightness Sync Data

Table 4-71 OSD BRIGHTNESS SYNC CMD DATA

| HEX  | ASCII | Operation |
|------|-------|-----------|
| 0x30 | '0'   | Off       |
| 0x31 | '1'   | On        |

## Trigger Data

Table 4-72 TRIGGER CMD DATA

| HEX  | ASCII | Operation |
|------|-------|-----------|
| 0x30 | '0'   | Off       |
| 0x31 | '1'   | Power     |
| 0x32 | '2'   | Anamo     |
| 0x33 | '3'   | Inst. 1   |
| 0x34 | '4'   | Inst. 2   |
| 0x35 | '5'   | Inst. 3   |
| 0x36 | '6'   | Inst. 4   |
| 0x37 | '7'   | Inst. 5   |

|      |     |          |
|------|-----|----------|
| 0x38 | '8' | Inst. 6  |
| 0x39 | '9' | Inst. 7  |
| 0x41 | 'A' | Inst. 8  |
| 0x42 | 'B' | Inst. 9  |
| 0x43 | 'C' | Inst. 10 |

## Off Timer Data

Table 4-73 OFF TIMER CMD DATA

| HEX  | ASCII | Operation |
|------|-------|-----------|
| 0x30 | '0'   | Off       |
| 0x31 | '1'   | 1 Hour    |
| 0x32 | '2'   | 2 Hours   |
| 0x33 | '3'   | 3 Hours   |
| 0x34 | '4'   | 4 Hours   |

## Eco Mode, HIDE Eco Data

Table 4-74 ECO MODE, HIDE ECO CMD DATA

| HEX  | ASCII | Operation |
|------|-------|-----------|
| 0x30 | '0'   | Off       |
| 0x31 | '1'   | On        |

## Control 4 Data

Table 4-75 CONTROL 4 CMD DATA

| HEX  | ASCII | Operation |
|------|-------|-----------|
| 0x30 | '0'   | Off       |
| 0x31 | '1'   | On        |

## Input Data

Table 4-76 INPUT CMD DATA

| HEX  | ASCII | Operation |
|------|-------|-----------|
| 0x36 | '6'   | HDMI-1    |
| 0x37 | '7'   | HDMI-2    |

## Source Data

Table 4-77 SOURCE CMD DATA

| Hex       | ASCII | Operation    | Hex       | ASCII | Operation         |
|-----------|-------|--------------|-----------|-------|-------------------|
| 0x30 0x32 | '0 2' | 480p         | 0x32 0x35 | '2 5' | VGA(640x480)      |
| 0x30 0x33 | '0 3' | 576p         | 0x32 0x36 | '2 6' | SVGA(800x600)     |
| 0x30 0x34 | '0 4' | 720p50       | 0x32 0x37 | '2 7' | XGA(1024x768)     |
| 0x30 0x35 | '0 5' | 720p60       | 0x32 0x38 | '2 8' | SXGA(1280x1024)   |
| 0x30 0x36 | '0 6' | 1080i50      | 0x32 0x39 | '2 9' | WXGA(1280x768)    |
| 0x30 0x37 | '0 7' | 1080i60      | 0x32 0x41 | '2 A' | WXGA+(1440x900)   |
| 0x30 0x38 | '0 8' | 1080p24      | 0x32 0x42 | '2 B' | WSXGA+(1680x1050) |
| 0x30 0x39 | '0 9' | 1080p50      | 0x32 0x43 | '2 C' | WUXGA(1920x1200)  |
| 0x30 0x41 | '0 A' | 1080p60      | 0x32 0x44 | '2 D' | WXGA(1280x800)    |
| 0x30 0x42 | '0 B' | No Signal    | 0x32 0x45 | '2 E' | FWXGA(1366x768)   |
| 0x30 0x43 | '0 C' | 720p 3D      | 0x32 0x46 | '2 F' | WXGA++(1600x900)  |
| 0x30 0x44 | '0 D' | 1080i 3D     | 0x33 0x30 | '3 0' | UXGA(1600x1200)   |
| 0x30 0x45 | '0 E' | 1080p 3D     | 0x33 0x31 | '3 1' | QXGA              |
| 0x30 0x46 | '0 F' | Out of Range | 0x33 0x32 | '3 2' | WQXGA             |

|           |       |                |           |       |                   |
|-----------|-------|----------------|-----------|-------|-------------------|
| 0x31 0x30 | '1 0' | 4K(4096)60     | 0x33 0x34 | '3 4' | 4096x2160 100Hz   |
| 0x31 0x31 | '1 1' | 4K(4096)50     | 0x33 0x35 | '3 5' | 3840x2160 100Hz   |
| 0x31 0x32 | '1 2' | 4K(4096)30     | 0x33 0x36 | '3 6' | 1080p100          |
| 0x31 0x33 | '1 3' | 4K(4096)25     | 0x33 0x37 | '3 7' | 1080p120          |
| 0x31 0x34 | '1 4' | 4K(4096)24     | 0x33 0x38 | '3 8' | 8K (7680x4320) 60 |
| 0x31 0x35 | '1 5' | 4K(3840)60     | 0x33 0x39 | '3 9' | 8K (7680x4320) 50 |
| 0x31 0x36 | '1 6' | 4K(3840)50     | 0x33 0x41 | '3 A' | 8K (7680x4320) 30 |
| 0x31 0x37 | '1 7' | 4K(3840)30     | 0x33 0x42 | '3 B' | 8K (7680x4320) 25 |
| 0x31 0x38 | '1 8' | 4K(3840)25     | 0x33 0x43 | '3 C' | 8K (7680x4320) 24 |
| 0x31 0x39 | '1 9' | 4K(3840)24     | 0x33 0x44 | '3 D' | WQHD60            |
| 0x31 0x43 | '1 C' | 1080p25        | 0x33 0x45 | '3 E' | WQHD120           |
| 0x31 0x44 | '1 D' | 1080p30        | 0x33 0x46 | '3 F' | 8K (7680x4320) 48 |
| 0x31 0x45 | '1 E' | 2048x1080 p24  |           |       |                   |
| 0x31 0x46 | '1 F' | 2048x1080 p25  |           |       |                   |
| 0x32 0x30 | '2 0' | 2048x1080 p30  |           |       |                   |
| 0x32 0x31 | '2 1' | 2048x1080 p50  |           |       |                   |
| 0x32 0x32 | '2 2' | 2048x1080 p60  |           |       |                   |
| 0x32 0x33 | '2 3' | 3840x2160 p120 |           |       |                   |
| 0x32 0x34 | '2 4' | 4096x2160 p120 |           |       |                   |

## Color Depth Data

Table 4-78 COLOR DEPTH CMD DATA

| HEX  | ASCII | Operation |
|------|-------|-----------|
| 0x30 | '0'   | 8 bit     |
| 0x31 | '1'   | 10 bit    |
| 0x32 | '2'   | 12 bit    |

## Color Space Data

Table 4-79 COLOR SPACE CMD DATA

| HEX  | ASCII | Operation |
|------|-------|-----------|
| 0x30 | '0'   | RGB       |
| 0x31 | '1'   | YCbCr422  |
| 0x32 | '2'   | YCbCr444  |
| 0x33 | '3'   | YCbCr420  |

## Colorimetry Data

Table 4-80 COLORIMETRY CMD DATA

| HEX  | ASCII | Operation                      |
|------|-------|--------------------------------|
| 0x30 | '0'   | No Data                        |
| 0x31 | '1'   | BT.601                         |
| 0x32 | '2'   | BT.709                         |
| 0x33 | '3'   | xvYCC601                       |
| 0x34 | '4'   | xvYCC709                       |
| 0x35 | '5'   | sYCC601                        |
| 0x36 | '6'   | Adobe YCC601                   |
| 0x37 | '7'   | Adobe RGB                      |
| 0x38 | '8'   | BT.2020 Constant Luminance     |
| 0x39 | '9'   | BT.2020 Non-Constant Luminance |
| 0x41 | 'A'   | sRGB                           |
| 0x42 | 'B'   | DCI-P3 (D65)                   |
| 0x43 | 'C'   | DCI-P3 (theater)               |

## HDR Data

Table 4-81 HDR CMD DATA

| HEX  | ASCII | Operation |
|------|-------|-----------|
| 0x30 | '0'   | SDR       |
| 0x31 | '1'   | HDR       |

|      |     |               |
|------|-----|---------------|
| 0x32 | '2' | SMPTE ST.2084 |
| 0x33 | '3' | Hybrid Log    |
| 0x34 | '4' | HDR10+        |
| 0x46 | 'F' | None          |

## DSC Mode Data

Table 4-82 DSC MODE CMD DATA

| HEX  | ASCII | Operation |
|------|-------|-----------|
| 0x30 | '0'   | DSC Off   |
| 0x31 | '1'   | DSC On    |

## Link Rate Data

Table 4-83 LINK RATE CMD DATA

| HEX  | ASCII | Operation            |
|------|-------|----------------------|
| 0x30 | '0'   | Disable ( TMDs モード ) |
| 0x31 | '1'   | 3 Gbps / 3 Lanes     |
| 0x33 | '3'   | 6 Gbps / 3 Lanes     |
| 0x34 | '4'   | 6 Gbps / 4 Lanes     |
| 0x35 | '5'   | 8 Gbps / 4 Lanes     |
| 0x36 | '6'   | 10 Gbps / 4 Lanes    |

### 4.10.2 Special 2 Data

This is the same as Gamma data [GammaRed, Green, Blue] of Gamma table "Custom 1/2/3".

### 4.10.3 Special 3 Data

This is the same as Model Status Asking [MoDel].

### 4.10.4 Special 4 Data

This is 2 Byte ASCII characters.

### 4.10.5 Special 10 Data

Parameter relating to editing names of Installation Mode 1/2/3/4/5/6/7/8/9/10 and User 1/2/3/4/5/6 in Picture Mode. 10 Byte ASCII characters. Communication format is same as Calibrator information transmission/display [PMCI].

### 4.10.6 Special 14 Data

Parameter relating to Software Version. 6 Byte ASCII characters. Front-align with 4-character version notation, fill in empty spaces with Space: 0x20.

## Operation

Use the command as shown in the following examples.

## (Example 1) Switching Picture Mode

**Purpose: To set Color Temp. Table to '6500K'**

(1) Transmitting Data from External controller to Projector as follows.

| 1: External Controller → |      |                |      |      |      |      |       |      |
|--------------------------|------|----------------|------|------|------|------|-------|------|
| 0x21                     | 0x89 | 0x01           | 0x50 | 0x4D | 0x43 | 0x4C | 0x32  | 0x0A |
| !' (Operation)           | PJ   | Individual : 1 | 'P'  | 'M'  | 'C'  | 'L'  | 6500K | End  |

(2) If the projector receives the data (1) above and the command receipt is normal, an ACK is returned from the projector as follows.

| 2: ← Projector |      |                |      |      |      |
|----------------|------|----------------|------|------|------|
| 0x06           | 0x89 | 0x01           | 0x50 | 0x4D | 0x0A |
| ACK            | PJ   | Individual : 1 | 'P'  | 'M'  | End  |

(3) The projector sets the Color Temp. Table to '6500K'.

## (Example 2) Contrast adjustment

**Purpose: To set Contrast to +20.**

(1) Transmit the data from the external controller to the projector as follows.

| 1: External controller → |      |                |      |      |      |      |      |      |      |      |      |
|--------------------------|------|----------------|------|------|------|------|------|------|------|------|------|
| 0x21                     | 0x89 | 0x01           | 0x50 | 0x4D | 0x43 | 0x4E | 0x30 | 0x30 | 0x31 | 0x34 | 0x0A |
| !' (Operation)           | PJ   | Individual : 1 | 'P'  | 'M'  | 'C'  | 'N'  | '0'  | '0'  | '1'  | '4'  | End  |

(2) If the projector receives the data (1) above and the command receipt is normal, an ACK is returned from the projector as follows.

| 2: ← Projector |      |                |      |      |      |
|----------------|------|----------------|------|------|------|
| 0x06           | 0x89 | 0x01           | 0x50 | 0x4D | 0x0A |
| ACK            | PJ   | Individual : 1 | 'P'  | 'M'  | End  |

(3) The projector sets the contrast to +20.

(Example 3) Gamma adjustment

**Purpose: To transmit gamma data of red to the projector.**

(1) Transmit the data from the external controller to the projector as follows.

| 1: External controller → |      |                |      |      |      |      |      |
|--------------------------|------|----------------|------|------|------|------|------|
| 0x21                     | 0x89 | 0x01           | 0x50 | 0x4D | 0x44 | 0x52 | 0x0A |
| '!(Operation)            | PJ   | Individual : 1 | 'P'  | 'M'  | 'D'  | 'R'  | End  |

(2) If the projector receives the data (1) above and the command receipt is normal, an ACK is returned from the projector as follows.

| 2: ← Projector |      |                |      |      |      |
|----------------|------|----------------|------|------|------|
| 0x06           | 0x89 | 0x01           | 0x50 | 0x4D | 0x0A |
| ACK            | PJ   | Individual : 1 | 'P'  | 'M'  | End  |

(3) The external controller sends 512 bytes binary data to the projector.

| 3: External controller → |  |
|--------------------------|--|
| 512 byte                 |  |
| Data parameter           |  |

(4) If the projector receives data (3) above and receipt was normal, an ACK is returned from the projector as follows.

| 4: ← Projector |      |                |      |      |      |
|----------------|------|----------------|------|------|------|
| 0x06           | 0x89 | 0x01           | 0x50 | 0x4D | 0x0A |
| ACK            | PJ   | Individual : 1 | 'P'  | 'M'  | End  |

## 4.10.7 Reference

Use the command as shown in the following examples.

(Example1) Confirm Picture Mode

**Purpose: To Confirm Current Picture Mode (Picture Mode: Natural)**

(1) Transmit the data from the external controller to the projector as follows.

| 1: External Controller → |      |                |      |      |      |      |      |
|--------------------------|------|----------------|------|------|------|------|------|
| 0x3F                     | 0x89 | 0x01           | 0x50 | 0x4D | 0x50 | 0x4D | 0x0A |
| '?(Reference)            | PJ   | Individual : 1 | 'P'  | 'M'  | 'P'  | 'M'  | End  |

(2) If the projector receives the data (1) above and the command receipt is normal, an ACK is returned from the projector as follows.

| 2: ← Projector |      |                |      |      |      |
|----------------|------|----------------|------|------|------|
| 0x06           | 0x89 | 0x01           | 0x50 | 0x4D | 0x0A |
| ACK            | PJ   | Individual : 1 | 'P'  | 'M'  | End  |

(3) Then, the projector transmits Picture Mode to the External Controller.

| 3: ← Projector |      |               |      |      |         |      |      |
|----------------|------|---------------|------|------|---------|------|------|
| 0x40           | 0x89 | 0x01          | 0x50 | 0x4D | 0x30    | 0x33 | 0x0A |
| '@(Reference)  | PJ   | Individual: 1 | 'P'  | 'M'  | Natural |      | End  |

By the above exchange, it can be confirmed that the projector's Picture Mode is 'Natural'.

(Example 2) Brightness confirmation

**Purpose: To confirm the brightness. (Current brightness: -3)**

(1) Transmit the data from the external controller to the projector as follows.

| 1: External controller → |      |                |      |      |      |      |      |
|--------------------------|------|----------------|------|------|------|------|------|
| 0x3F                     | 0x89 | 0x01           | 0x50 | 0x4D | 0x42 | 0x52 | 0x0A |
| '?(Reference)            | PJ   | Individual : 1 | 'P'  | 'M'  | 'B'  | 'R'  | End  |

(2) If the projector receives the data (1) above and the command receipt is normal, an ACK is returned from the projector as follows.

| 2: ← Projector |      |                |      |      |      |
|----------------|------|----------------|------|------|------|
| 0x06           | 0x89 | 0x01           | 0x50 | 0x4D | 0x0A |
| ACK            | PJ   | Individual : 1 | 'P'  | 'M'  | End  |

(3) The projector transmits brightness setting “-3” to the external controller.

| 3: ← Projector |      |                |      |      |      |      |      |      |      |
|----------------|------|----------------|------|------|------|------|------|------|------|
| 0x40           | 0x89 | 0x01           | 0x50 | 0x4D | 0x46 | 0x46 | 0x46 | 0x44 | 0x0A |
| '@(Response)   | PJ   | Individual : 1 | 'P'  | 'M'  | 'F'  | 'F'  | 'F'  | 'D'  | End  |

By the above exchange, it can be confirmed that the projector's brightness is set to “-3”.

(Example 3) Gamma confirmation

### Purpose: To confirm the Green gamma data.

(1) Transmit the data from the external controller to the projector as follows.

| 1: External controller → |      |                |      |      |      |      |      |
|--------------------------|------|----------------|------|------|------|------|------|
| 0x3F                     | 0x89 | 0x01           | 0x50 | 0x4D | 0x44 | 0x47 | 0x0A |
| '?(Reference)            | PJ   | Individual : 1 | 'P'  | 'M'  | 'D'  | 'G'  | End  |

(2) If the projector receives the data (1) above and the command receipt is normal, an ACK is returned from the projector as follows.

| 2: ← Projector |      |                |      |      |      |
|----------------|------|----------------|------|------|------|
| 0x06           | 0x89 | 0x01           | 0x50 | 0x4D | 0x0A |
| ACK            | PJ   | Individual : 1 | 'P'  | 'M'  | End  |

(3) The projector transmits 512 bytes binary data to the external controller.

(Example 4) Confirm Software Version

### Purpose: Confirm Software Version

(1) Transmit the data from the external controller to the projector as follows.

| 1: External Controller → |      |              |      |      |      |      |      |
|--------------------------|------|--------------|------|------|------|------|------|
| 0x3F                     | 0x89 | 0x01         | 0x49 | 0x46 | 0x53 | 0x56 | 0x0A |
| '?'<br>(Reference)       | PJ   | Individual:1 | 'I'  | 'F'  | 'S'  | 'V'  | End  |

(2) If the projector receives the data (1) above and the command receipt is normal, an ACK is returned from the projector as follows.

| 2: ←Projector |      |                |      |      |      |
|---------------|------|----------------|------|------|------|
| 0x06          | 0x89 | 0x01           | 0x49 | 0x46 | 0x0A |
| ACK           | PJ   | Individual : 1 | 'I'  | 'F'  | End  |

(3) Then, the Projector transmits Software version to the external controller.

| 3: ←Projector |      |                |      |      |      |      |      |      |      |      |      |
|---------------|------|----------------|------|------|------|------|------|------|------|------|------|
| 0x06          | 0x89 | 0x01           | 0x49 | 0x46 | 0x30 | 0x33 | 0x2D | 0x30 | 0x30 | 0x35 | 0x0A |
| ACK           | PJ   | Individual : 1 | 'I'  | 'F'  | '0'  | '3'  | '-'  | '0'  | '0'  | '5'  | End  |

By the above exchange, it can be confirmed that the Software version is "03.005".



## 4.11 LAN setup [Lan Setup]

Used to setup LAN configuration.

### Command code

Table 4-84 LAN SETUP CMD

| HEX  |      | ASCII |     | Function              |
|------|------|-------|-----|-----------------------|
| 0x4C | 0x53 | 'L'   | 'S' | LAN setup [Lan Setup] |

### Parameter1

Data length: No regulation

Sub command table (Mandatory command only)

Table 4-85 LAN SETUP CMD SUB

| HEX  |      | ASCII |     | Function                | Last memory | Operation                         | Reference |
|------|------|-------|-----|-------------------------|-------------|-----------------------------------|-----------|
| 0x44 | 0x53 | 'D'   | 'S' | DHCP Client setting     | Yes         | ✓                                 | ✓         |
| 0x49 | 0x50 | 'I'   | 'P' | IP Address setting      | Yes         | ✓<br>(When 'DHCP Client' is Off.) | ✓         |
| 0x53 | 0x4D | 'S'   | 'M' | Subnet Mask setting     | Yes         | ✓<br>(When 'DHCP Client' is Off.) | ✓         |
| 0x44 | 0x47 | 'D'   | 'G' | Default Gateway setting | Yes         | ✓<br>(When 'DHCP Client' is Off.) | ✓         |
| 0x4D | 0x41 | 'M'   | 'A' | MAC Address setting     | Yes         | -                                 | ✓         |
| 0x52 | 0x53 | 'R'   | 'S' | Network reboot          | No          | ✓                                 | -         |
| 0x50 | 0x54 | 'P'   | 'T' | Port setting            | Yes         | ✓                                 | ✓         |

- It consists of "Sub command" + "setting". Sub command consists of ASCII character 2 bytes.

### Parameter2

Sub command parameters are as follows.

Parameters when Sub command is [0x44,0x53]: Data length 1

Table 4-86 DHCP Client

| HEX  | ASCII | Operation   |
|------|-------|-------------|
| 0x30 | '0'   | Off(Static) |
| 0x31 | '1'   | On          |

Parameters when Sub command is [0x49,0x50] [0x53,0x4D] [0x44,0x47]: Data length 8

Parameters when Sub command is [0x4D,0x41]: Data length 12

Parameters when Sub command is [0x52,0x53]: Data length 1

Table 4-87 NETWORK RESTART

| HEX  | ASCII | Operation       |
|------|-------|-----------------|
| 0x31 | '1'   | Network Restart |

Parameters when Sub command is [0x50,0x54]: Data length 4

### 4.11.1 Operation

Use the command as shown in the following examples.

(Example1) DHCP Client setting

#### Purpose: To set DHCP Client to On.

(1) Transmit the data from the external controller to the projector as follows.

| 1: External controller → |      |                |      |      |      |      |      |      |
|--------------------------|------|----------------|------|------|------|------|------|------|
| 0x21                     | 0x89 | 0x01           | 0x4C | 0x53 | 0x44 | 0x53 | 0x31 | 0x0A |
| !(Operation)             | PJ   | Individual : 1 | 'L'  | 'S'  | 'D'  | 'S'  | On   | End  |

(2) If the projector receives the data (1) above and the command receipt is normal, an ACK is returned from the projector as follows.

| 2: ← Projector |      |                |      |      |      |
|----------------|------|----------------|------|------|------|
| 0x06           | 0x89 | 0x01           | 0x4C | 0x53 | 0x0A |
| ACK            | PJ   | Individual : 1 | 'L'  | 'S'  | End  |

(3) The projector set DHCP Client to On.

(Example2) IP Address setting

#### Purpose: To set IP Address to 192.168.1.10.

(1) Transmit the data from the external controller to the projector as follows.

| 1: External controller → |      |                |      |      |      |      |            |            |          |           |      |      |      |      |
|--------------------------|------|----------------|------|------|------|------|------------|------------|----------|-----------|------|------|------|------|
| 0x21                     | 0x89 | 0x01           | 0x4C | 0x53 | 0x49 | 0x50 | 0x43       | 0x30       | 0x41     | 0x38      | 0x30 | 0x31 | 0x30 | 0x41 |
| !(Operation)             | PJ   | Individual : 1 | 'L'  | 'S'  | 'I'  | 'P'  | 192(=0xC0) | 168(=0xA8) | 1(=0x01) | 10(=0x0A) |      |      |      | End  |

(2) If the projector receives the data (1) above and the command receipt is normal, an ACK is returned from the projector as follows.

| 2: ← Projector |      |                |      |      |      |
|----------------|------|----------------|------|------|------|
| 0x06           | 0x89 | 0x01           | 0x4C | 0x53 | 0x0A |
| ACK            | PJ   | Individual : 1 | 'L'  | 'S'  | End  |

(3) The projector saves "192.168.1.10." to IP Address. (Valid after network reboot)

(Example3) Network reboot

#### To reboot Network

(1) Transmit the data from the external controller to the projector as follows.

| 1: External controller → |      |                |      |      |      |      |         |      |
|--------------------------|------|----------------|------|------|------|------|---------|------|
| 0x21                     | 0x89 | 0x01           | 0x4C | 0x53 | 0x52 | 0x53 | 0x31    | 0x0A |
| !(Operation)             | PJ   | Individual : 1 | 'L'  | 'S'  | 'R'  | 'S'  | Execute | End  |

(2) If the projector receives the data (1) above and the command receipt is normal, an ACK is returned from the projector as follows.

| 2: ← Projector |      |                |      |      |      |
|----------------|------|----------------|------|------|------|
| 0x06           | 0x89 | 0x01           | 0x4C | 0x53 | 0x0A |
| ACK            | PJ   | Individual : 1 | 'L'  | 'S'  | End  |

(3) The projector reboots the Network.

(Example4) Port setting

**Purpose: To set the Port to 10000(=0x2710).**

(1) Transmit the data from the external controller to the projector as follows.

| 1: External controller → |      |                |      |      |      |      |                |      |      |      |      |
|--------------------------|------|----------------|------|------|------|------|----------------|------|------|------|------|
| 0x21                     | 0x89 | 0x01           | 0x4C | 0x53 | 0x50 | 0x54 | 0x32           | 0x37 | 0x31 | 0x30 | 0x0A |
| '!(Operation)            | PJ   | Individual : 1 | 'L'  | 'S'  | 'P'  | 'T'  | 10000(=0x2710) |      |      |      | End  |

(2) If the projector receives the data (1) above and the command receipt is normal, an ACK is returned from the projector as follows.

| 2: ← Projector |      |                |      |      |      |
|----------------|------|----------------|------|------|------|
| 0x06           | 0x89 | 0x01           | 0x4C | 0x53 | 0x0A |
| ACK            | PJ   | Individual : 1 | 'L'  | 'S'  | End  |

(3) The projector saves 10000 to the Port. (No matter what the Network reboot)

## 4.11.2 Reference

Use the command as shown in the following examples.

(Example1) DHCP Client confirmation

**Purpose: To confirm DHCP Client.**

(1) Transmit the data from the external controller to the projector as follows.

| 1: External controller → |      |                |      |      |      |      |      |
|--------------------------|------|----------------|------|------|------|------|------|
| 0x3F                     | 0x89 | 0x01           | 0x4C | 0x53 | 0x44 | 0x53 | 0x0A |
| '?(Reference)            | PJ   | Individual : 1 | 'L'  | 'S'  | 'D'  | 'S'  | End  |

(2) If the projector receives the data (1) above and the command receipt is normal, an ACK is returned from the projector as follows.

| 2: ← Projector |      |                |      |      |      |
|----------------|------|----------------|------|------|------|
| 0x06           | 0x89 | 0x01           | 0x4C | 0x53 | 0x0A |
| ACK            | PJ   | Individual : 1 | 'L'  | 'S'  | End  |

(3) The projector transmits DHCP Client to the external controller.

| 3: ← Projector |      |                |      |      |      |      |
|----------------|------|----------------|------|------|------|------|
| 0x40           | 0x89 | 0x01           | 0x4C | 0x53 | 0x30 | 0x0A |
| '@(Response)   | PJ   | Individual : 1 | 'L'  | 'S'  | Off  | End  |

By the exchange above, it can be confirmed that the projector DHCP Client is set to Off.

(Example2) MAC Address confirmation

**Purpose: To confirm the current MAC Address.**

(1) Transmit the data from the external controller to the projector as follows.

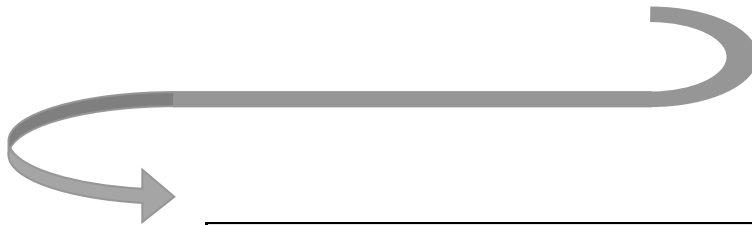
| 1: External controller → |      |                |      |      |      |      |      |
|--------------------------|------|----------------|------|------|------|------|------|
| 0x3F                     | 0x89 | 0x01           | 0x4C | 0x53 | 0x4D | 0x41 | 0x0A |
| '?(Reference)            | PJ   | Individual : 1 | 'L'  | 'S'  | 'M'  | 'A'  | End  |

(2) If the projector receives the data (1) above and the command receipt is normal, an ACK is returned from the projector as follows.

| 2: ← Projector |      |                |      |      |      |
|----------------|------|----------------|------|------|------|
| 0x06           | 0x89 | 0x01           | 0x4C | 0x53 | 0x0A |
| ACK            | PJ   | Individual : 1 | 'L'  | 'S'  | End  |

(3) The projector transmits MAC Address to the external controller.

| 3: ← Projector |      |                |      |      |
|----------------|------|----------------|------|------|
| 0x40           | 0x89 | 0x01           | 0x4C | 0x53 |
| '@(Response)   | PJ   | Individual : 1 | 'L'  | 'S'  |



|      |      |      |      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|------|------|------|
| 0x45 | 0x30 | 0x44 | 0x41 | 0x44 | 0x43 | 0x31 | 0x32 | 0x33 | 0x34 | 0x35 | 0x36 | 0x0A |
| E0   | DA   | DC   | 12   | 34   | 56   | End  |      |      |      |      |      |      |

By the above exchange, it can be confirmed that the projector's MAC Address is set to "E0:DA:DC:12:34:56".

(Example3) Port confirmation

**Purpose: To confirm the current Port.**

(1) Transmit the data from the external controller to the projector as follows.

| 1: External controller → |      |                |      |      |      |      |      |
|--------------------------|------|----------------|------|------|------|------|------|
| 0x3F                     | 0x89 | 0x01           | 0x4C | 0x53 | 0x50 | 0x54 | 0x0A |
| '?(Reference)            | PJ   | Individual : 1 | 'L'  | 'S'  | 'P'  | 'T'  | End  |

(2) If the projector receives the data (1) above and the command receipt is normal, an ACK is returned from the projector as follows.

| 2: ← Projector |      |                |      |      |      |
|----------------|------|----------------|------|------|------|
| 0x06           | 0x89 | 0x01           | 0x4C | 0x53 | 0x0A |
| ACK            | PJ   | Individual : 1 | 'L'  | 'S'  | End  |

(3) The projector transmits Port to the external controller.

| 3: ← Projector |      |                |      |      |              |      |      |      |      |
|----------------|------|----------------|------|------|--------------|------|------|------|------|
| 0x40           | 0x89 | 0x01           | 0x4C | 0x53 | 0x35         | 0x30 | 0x34 | 0x41 | 0x0A |
| '@(Response)   | PJ   | Individual : 1 | 'L'  | 'S'  | 20554(=504A) |      |      |      | End  |

By the above exchange, it can be confirmed that the projector's port is set to 20554(=0x504A).

End of specification